Chapter 3: RF Amplifier

3.1. Introduction

This tutorial demonstrates how to model an RF amplifier using Ansys Icepak as well as many features and functions essential to any Ansys Icepak project. For the sake of brevity, many of the later tutorials do not cover basic steps or explain the steps in detail as those tutorials assume you have completed this and/or the last tutorial (Finned Heat Sink (p. 5)) beforehand.

In this tutorial you will learn how to:

- Create a new project.
- Create a model using openings, fans, sources, enclosures, PCBs, heat sinks, and walls.
- Use non-conformal meshing.
- Set up a simulation with various physical conditions and parameters, including turbulence and natural convection.
- Calculate a solution.
- Post-process your results by using power and temperature limits, object faces, plane cuts, isosurfaces and variation plots.

Note:

Mesh counts and solution results may differ slightly due to enhancements to the meshing algorithm and Fluent solver, respectively.

3.2. Prerequisites

This tutorial assumes that you have little experience with Ansys Icepak but that you are generally familiar with the interface. If you are not, review the Sample Session in the Icepak User's Guide.

3.3. Problem Description

RF amplifiers are typically sealed enclosures that are placed within larger systems. They present a challenge from the thermal management perspective because no direct exchange of air exists between the interior of the amplifier and the ambient. The common method of cooling such subsystems is to mount a large heat sink on the amplifier housing that cools all the devices within the enclosure. A simplified version of an RF amplifier (Figure 3.1: Schematic of the RF Amplifier (p. 52)) will serve as the model for

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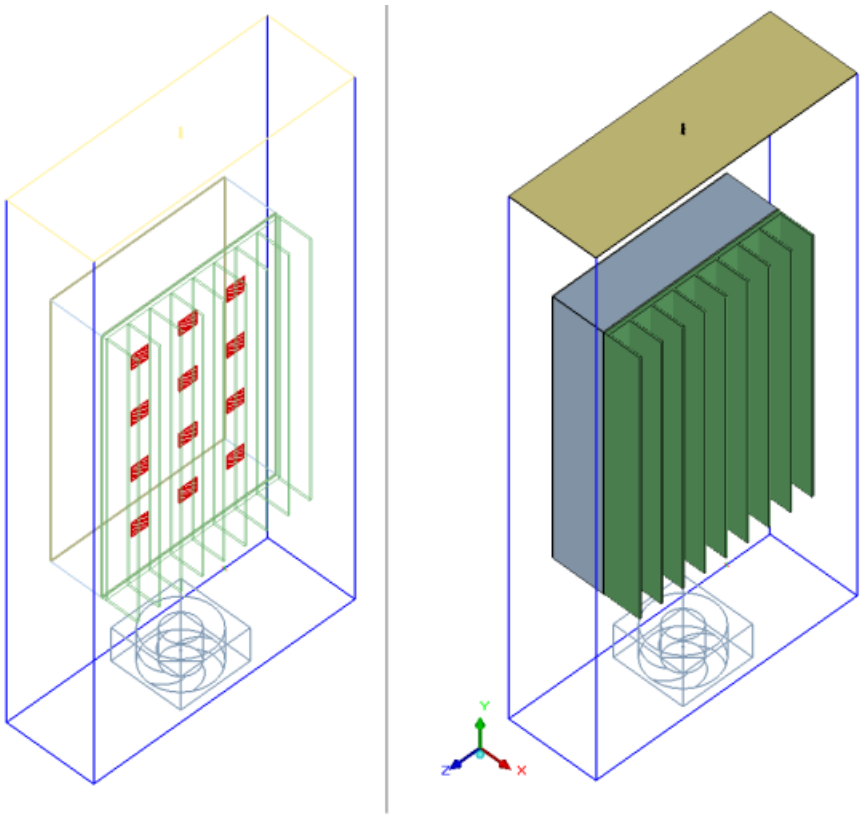
this tutorial. There will be free convection inside the amplifier and forced convection in the external domain.

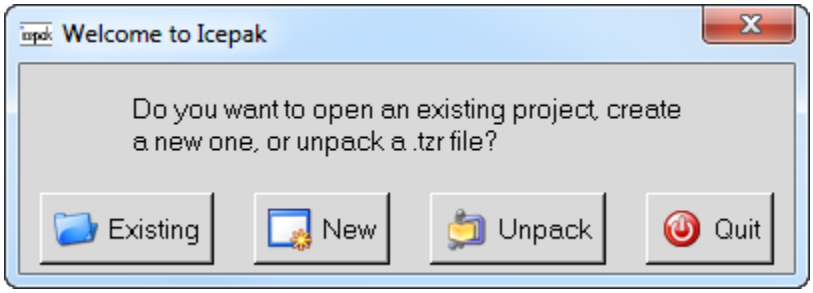
Figure 3.1: Schematic of the RF Amplifier



3.4. Step 1: Create a New Project

1. Start Ansys Icepak, as described in Starting Ansys Icepak from the User's Guide.

   When Ansys Icepak starts, the **Welcome to Icepak** panel opens automatically.



2. Click **New** in the **Welcome to Icepak** panel to start a new Ansys Icepak project.

   The **New project** panel appears.

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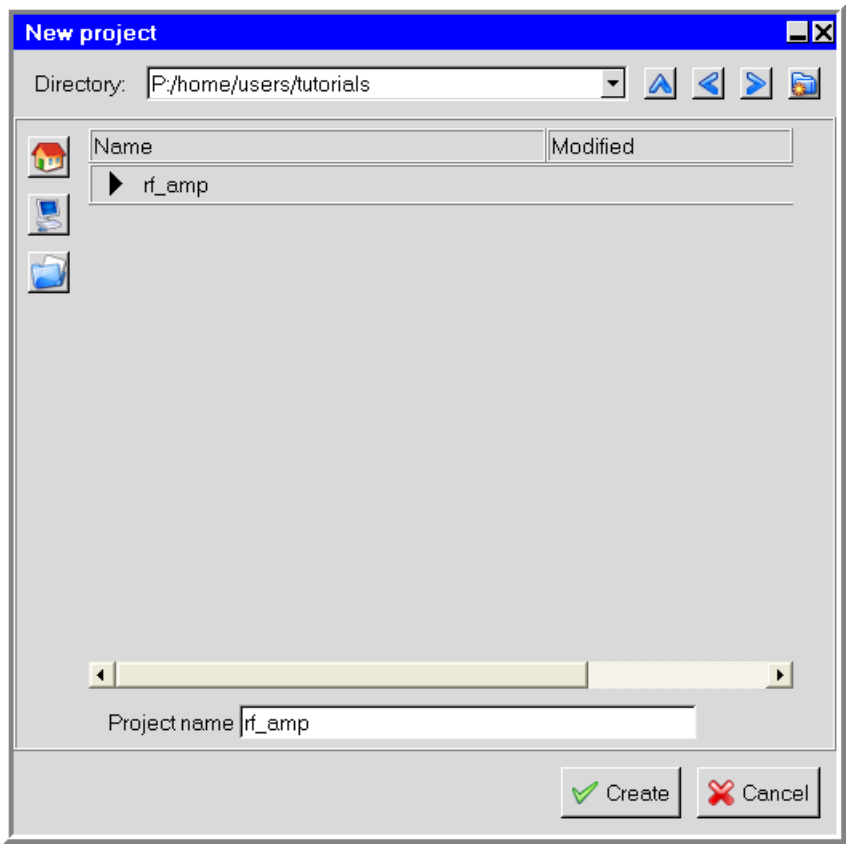

3. Specify the name **`rf_amp`** for your project and click **Create**.

   Ansys Icepak creates a default cabinet with the dimensions 1 m × 1 m × 1 m, and displays the cabinet in the graphics window.

Note:

You can rotate the cabinet around a central point using the left mouse button, or you can translate it to any point on the screen using the middle mouse button. You can zoom into and out from the cabinet using the right mouse button. To restore the cabinet to its default orientation, select the **Home position** in the **Orient** menu. Alternatively, you can click the **Home position** icon ( ) above the graphics window or press the **H** key.

3.5. Step 2: Build the Model

To build the model, you will first resize the cabinet to its proper size. Then you will create the amplifier housing, devices (heat sources), PCB, heatsink, fan and other geometrical objects.

1. Resize the default cabinet and create an opening on one side of the cabinet.

   Model → **Cabinet**

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Select the cabinet in the **Model manager** window and specify the following in the object geometry window:

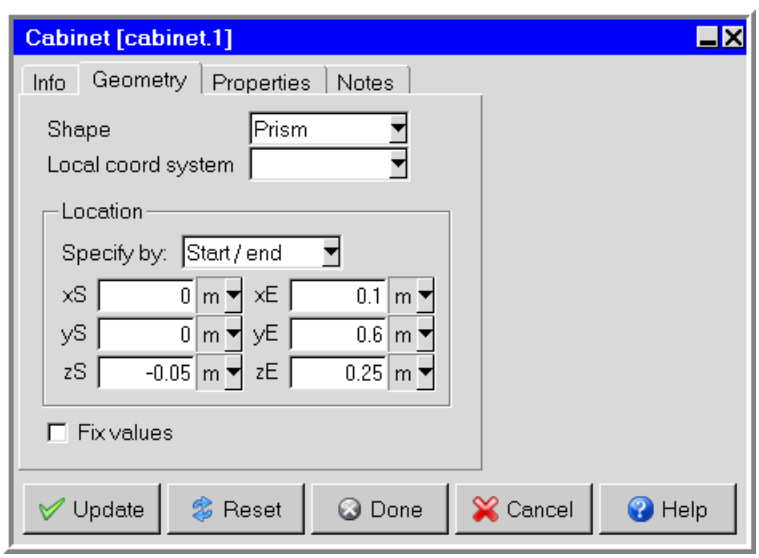
xS	0	m	xE	0.1	m
yS	0	m	yE	0.6	m
zS	-0.05	m	zE	0.25	m

Note:

After selecting the object to be edited in the **Model manager** window, there are several ways you can open the **Edit** panel:

- Double-click the object in the **Model manager** window, or
  - Type **Ctrl**+**e**, or
  - Right-click the object in the **Model manager** window and scroll to **Edit** object, or
  - Click the **Edit** button in the object geometry window, or
  - Click the **Edit object** icon ( ) in the model toolbar

Figure 3.2: The Cabinet Geometry Tab Panel



One side of this cabinet has an opening. To assign **Properties** on this boundary, in the **Properties** tab of the **Cabinet** object panel (Figure 3.3: The Cabinet Boundary Panel (p. 55)):

a. Change the **Max y Wall type** to be an **Opening**.

b. Click **Done** to accept the inputs and close the panel.

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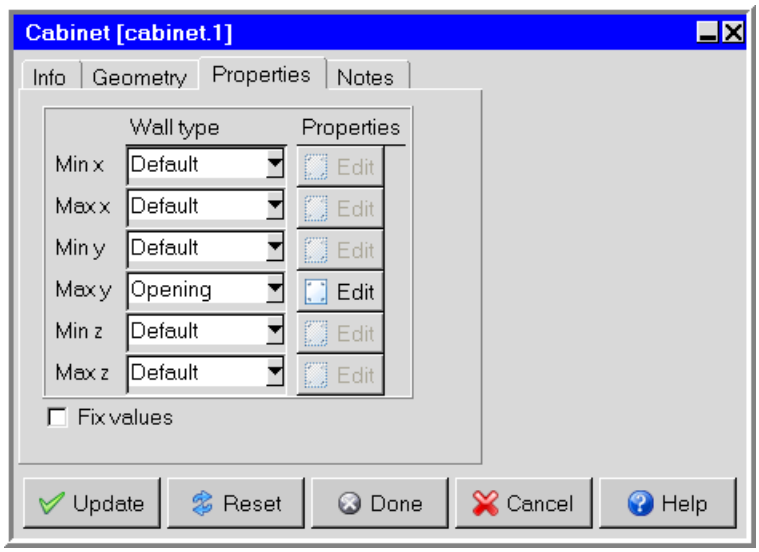
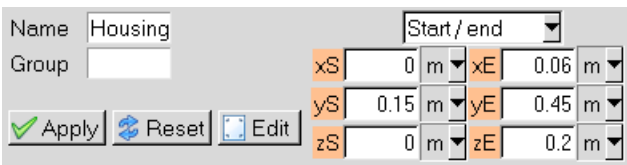
Figure 3.3: The Cabinet Boundary Panel



2. Create the Y and Z faces of the amplifier housing as an enclosure using the enclosure object.

   Click the **Create enclosures** icon ( ) in the model toolbar, then specify the following **Name** and dimensions:



   In the **Properties** tab specify the followings:

   a. Change the **Boundary type** to **Open** for **Min X** and **Max X**. For others, retain the boundary type as **Thin**.

   b. Specify the **Solid material** as **Polystyrene-rigid-R12**.

   > **Tip:**
   >
   > You have to scroll down the list to find this material.

   c. Click **Done**.

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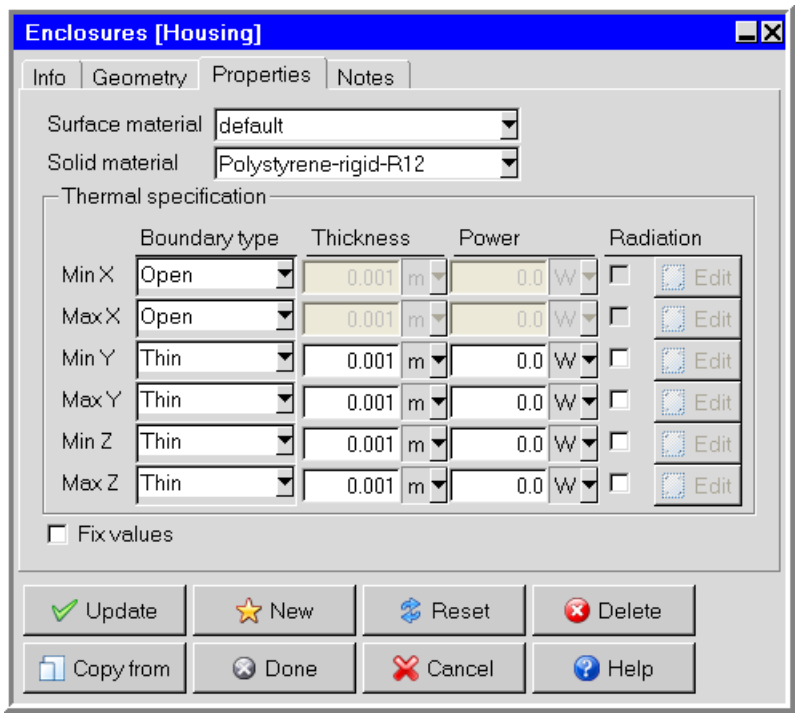
Figure 3.4: The Enclosure Panel



3. Create a wall on the Xmin face of the amplifier housing to cover the Xmin side of the enclosure. Click the **Create walls** icon ( ) in the model toolbar to create a new wall.

   In the object edit window, name the wall `Xmin` and change the plane to **Y-Z**.

4. Use **Morph Edges** to align the wall and then specify the wall properties.

Note:

Although you are using the align tools to place the wall at the desired locations, you could also specify the dimensions/locations of the wall in the **Geometry** tab and achieve the same result. However, the align tools are faster and thus the recommended method.

To start the process, click the **Morph edges** icon ( ) in the model toolbar. Now, follow the step-by-step procedure described below:

a. Select the Zmax edge of the wall, indicated by the red edge in the figure (Figure 3.5: Schematic Showing Edge Identities for Alignment (p. 57)) by left mouse clicking it in the graphical window. Notice that it turns red to indicate that it has been selected.

b. Click the middle mouse button to accept this edge.

c. Select the lower Zmax edge of the enclosure, indicated by the yellow edge in the figure (Figure 3.5: Schematic Showing Edge Identities for Alignment (p. 57)) with the left mouse button. Notice that it turns yellow to indicate its selection.

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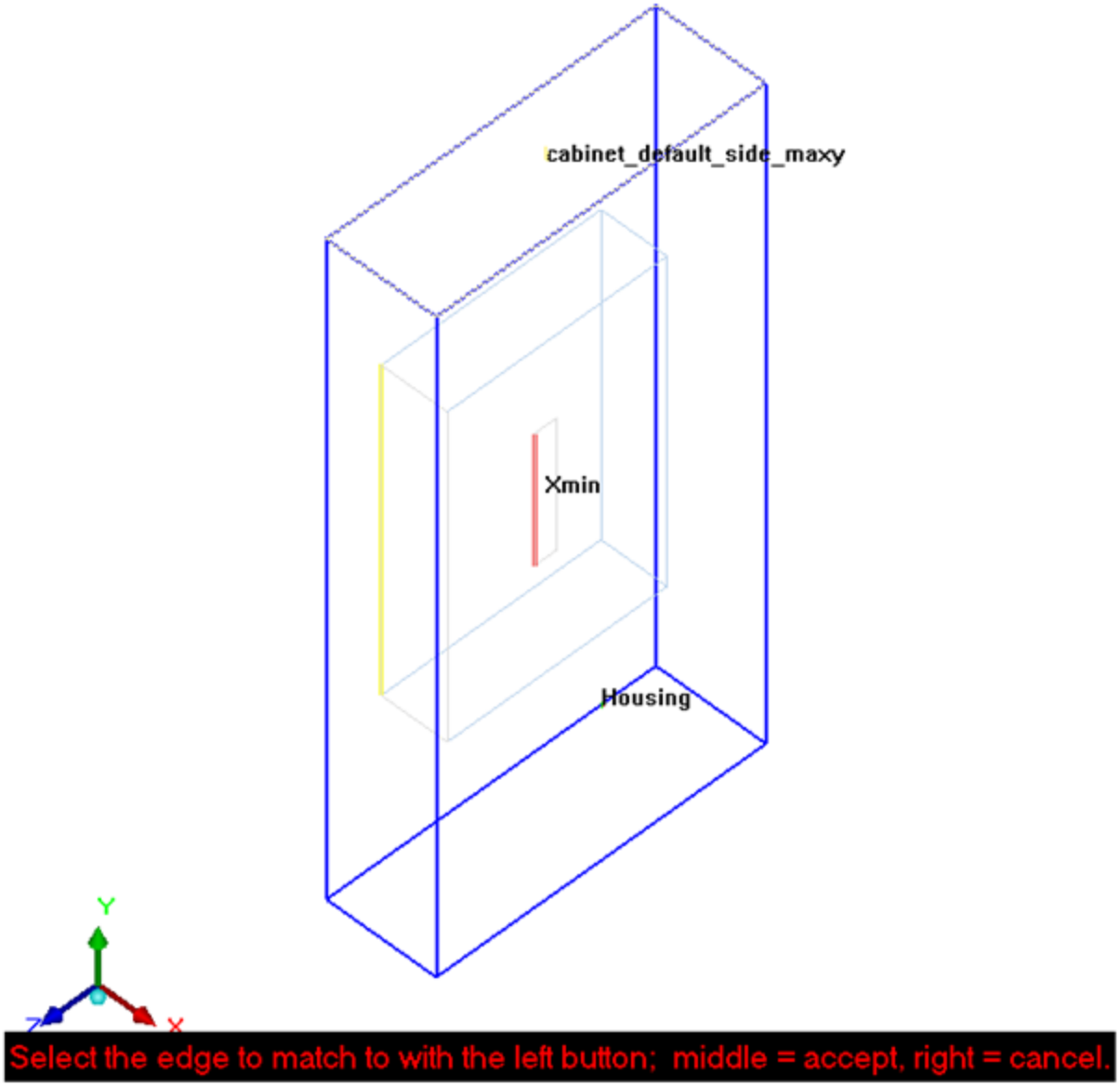
Figure 3.5: Schematic Showing Edge Identities for Alignment



d. Click the middle mouse button to accept the transformation. The wall **Xmin** should have now been moved and resized. Now the wall should extend to the entire Xmin side of the enclosure.

To specify the remaining wall dimension, stay in the **Morph edges** mode and complete the following steps:

a. Click the **Zmin** edge of the wall with the left mouse button. Be sure that the **Zmin** edge of the wall (and not the enclosure edge) is highlighted in red. By repeatedly clicking the left mouse button, Ansys Icepak cycles through all possible edges.

b. Click the middle mouse button to accept.

c. Using the left mouse button, click the lower Zmin edge of the enclosure.

d. Click the middle mouse button to accept. The wall now forms the Xmin face of the enclosure.

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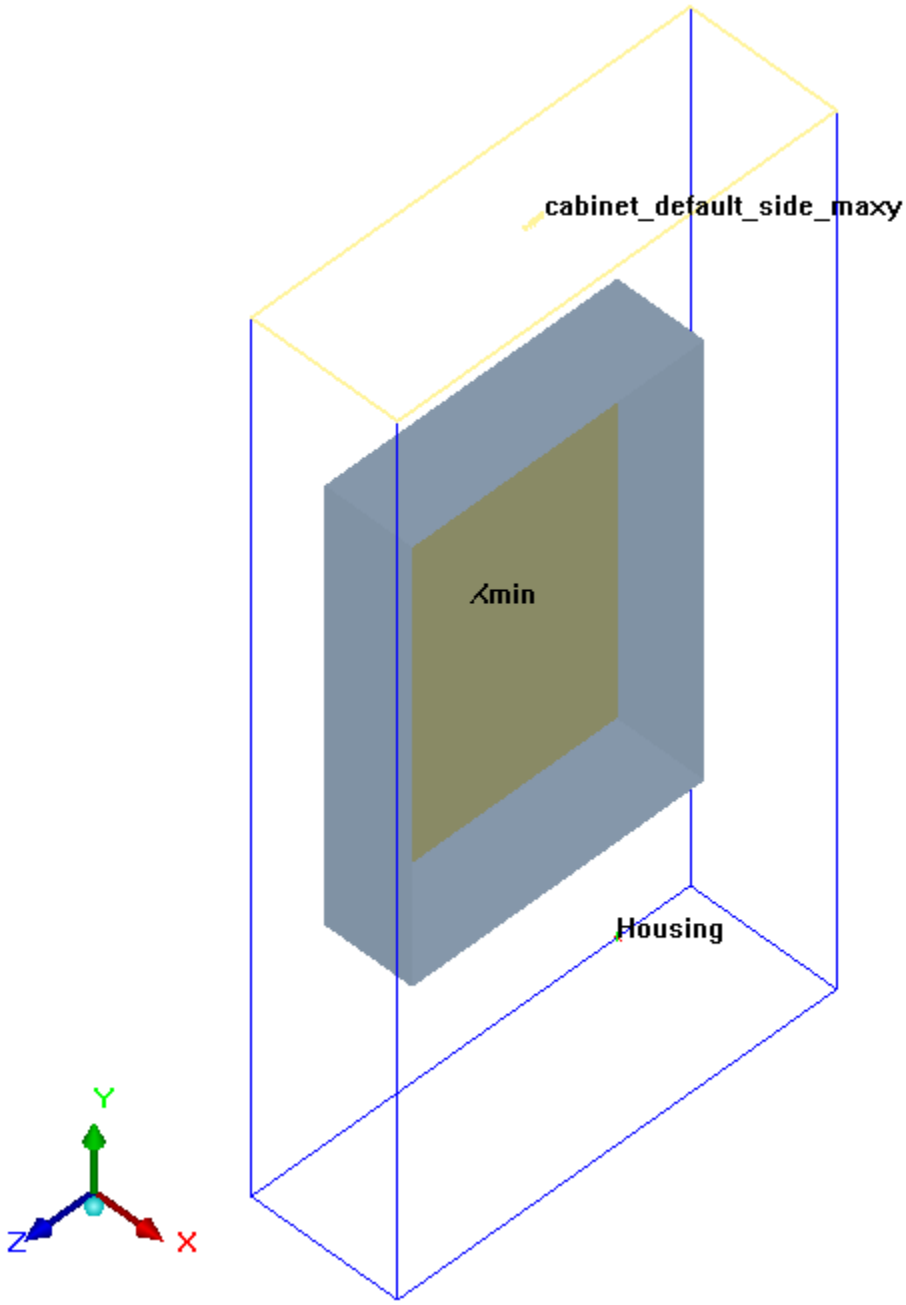
e. Click the right mouse button to exit the **Morph edges** mode.

The resulting model is shown in Figure 3.6: Geometry with Wall (p. 58) with shading to highlight new definitions. Shading is available under the **Info** tab in most panels.

Figure 3.6: Geometry with Wall



Double-click the newly created wall object (**Xmin**) in the **Model manager** window to open the **Walls** panel. Specify the following properties to the wall in the **Properties** tab.

a. Specify a **Wall thickness** of **`1`** mm (**`0.001`** m).

b. Specify the **Solid material** as **Polystyrene-rigid-R12** under **Plastics**.

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c. Specify the **External conditions** as **Heat transfer coefficient** and click the **Edit** button.

The **Wall external thermal conditions** panel opens.

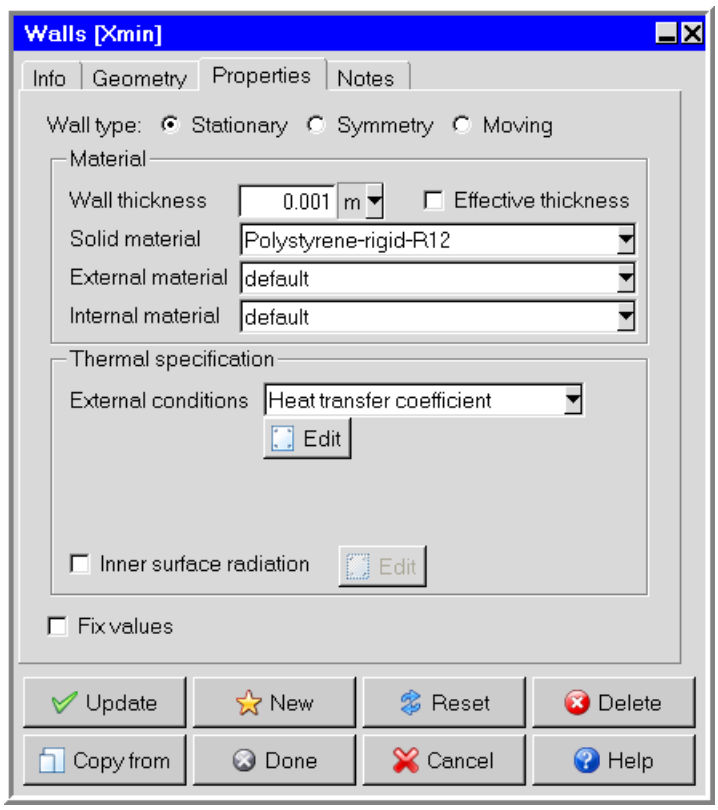
i. Select **Heat transfer coeff** in the **Thermal conditions** group box.

ii. Set the **Heat transfer coeff** to 5 W/K-m2.

iii. Click **Done** to close the **Wall external thermal conditions** panel.

iv. Click **Done** to close **Walls** panel (Figure 3.7: The Walls Panel (p. 59)).

Figure 3.7: The Walls Panel

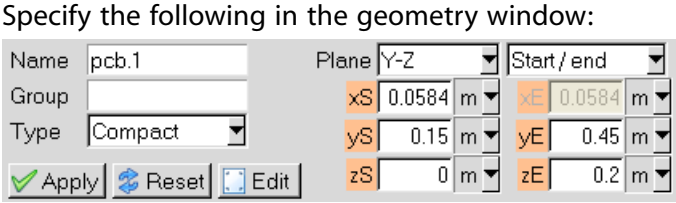


5. Create the PCB.

The PCB will cover the **Xmax** side of the enclosure.

a. Click the **Create printed circuit boards** icon ( ) in the **Model toolbar** to create a PCB and double-click the PCB object in the **Model manager** window.

b. Specify the following in the geometry window:



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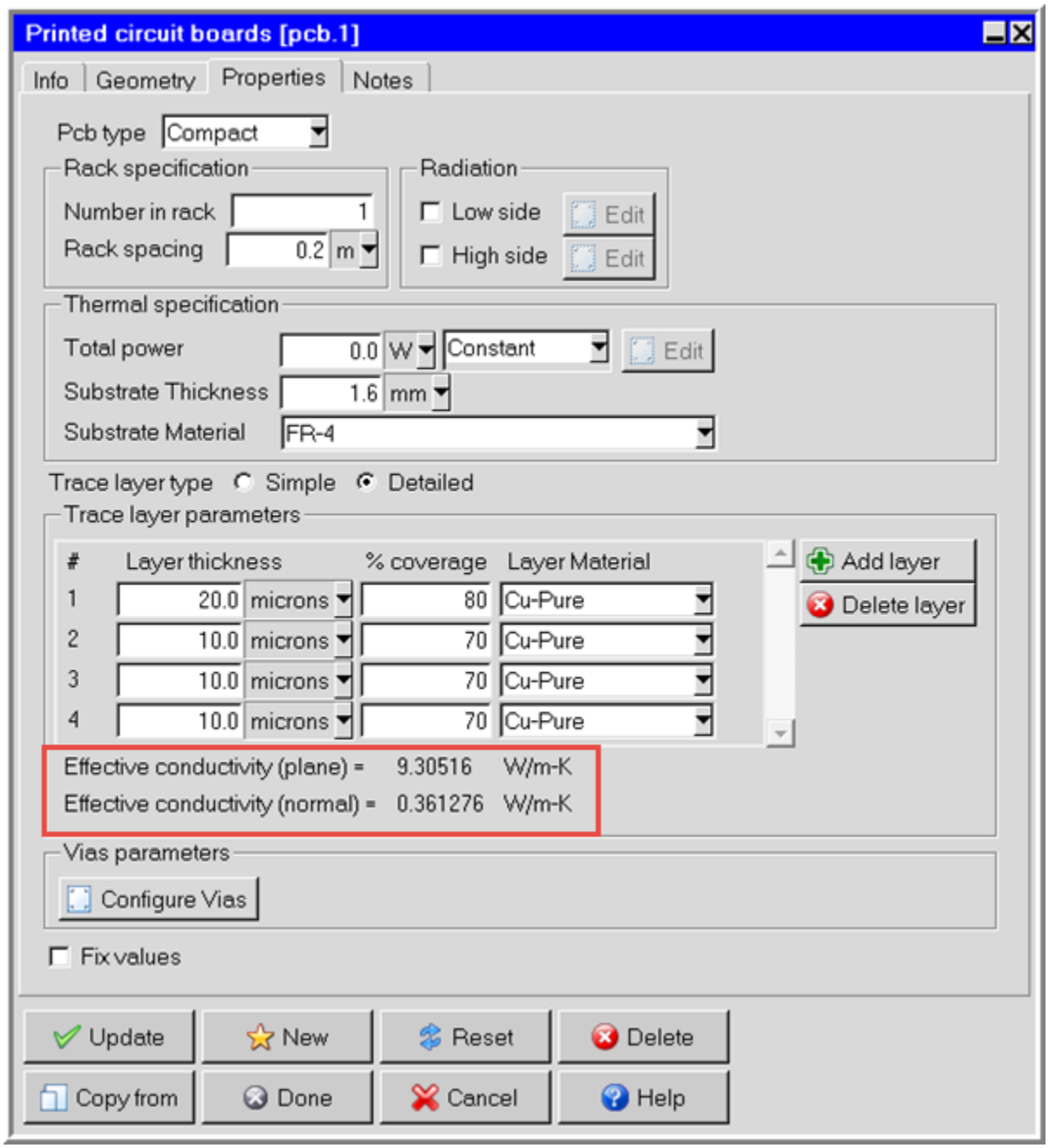
c. Specify the **Trace layer type** as **Detailed** and specify the parameters under **Trace layer parameters** (make sure that you enter both columns) in the **Properties** tab as shown in Figure 3.8: The Printed circuit boards Panel (p. 60). Click the **Add layer** button 3 more times. There is a total of four internal layers.

Note:

Specify the layer thicknesses in microns.

Notice that the **Effective conductivity** in plane and normal directions are updated when you click the **Update** button (Figure 3.8: The Printed circuit boards Panel (p. 60)).

Figure 3.8: The Printed circuit boards Panel



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d. Click **Done** to close the **Printed circuit boards** panel.

6. Create the devices as 2D sources.

   There are 12 devices on the bottom side of the PCB. Create these devices as 2D sources. The following steps show you how to create one and then use the copy utility to create the remaining 11 sources.

   a. Click the **Create sources** icon ( ) in the model toolbar to create a source and double-click the source object in the **Model manager** window.

   b. Specify the following source name, dimensions, and properties.

   c. In the **Properties** tab, specify the **Total power** as **7** W (Figure 3.9: The Sources Panel (p. 62)) and click **Done**.

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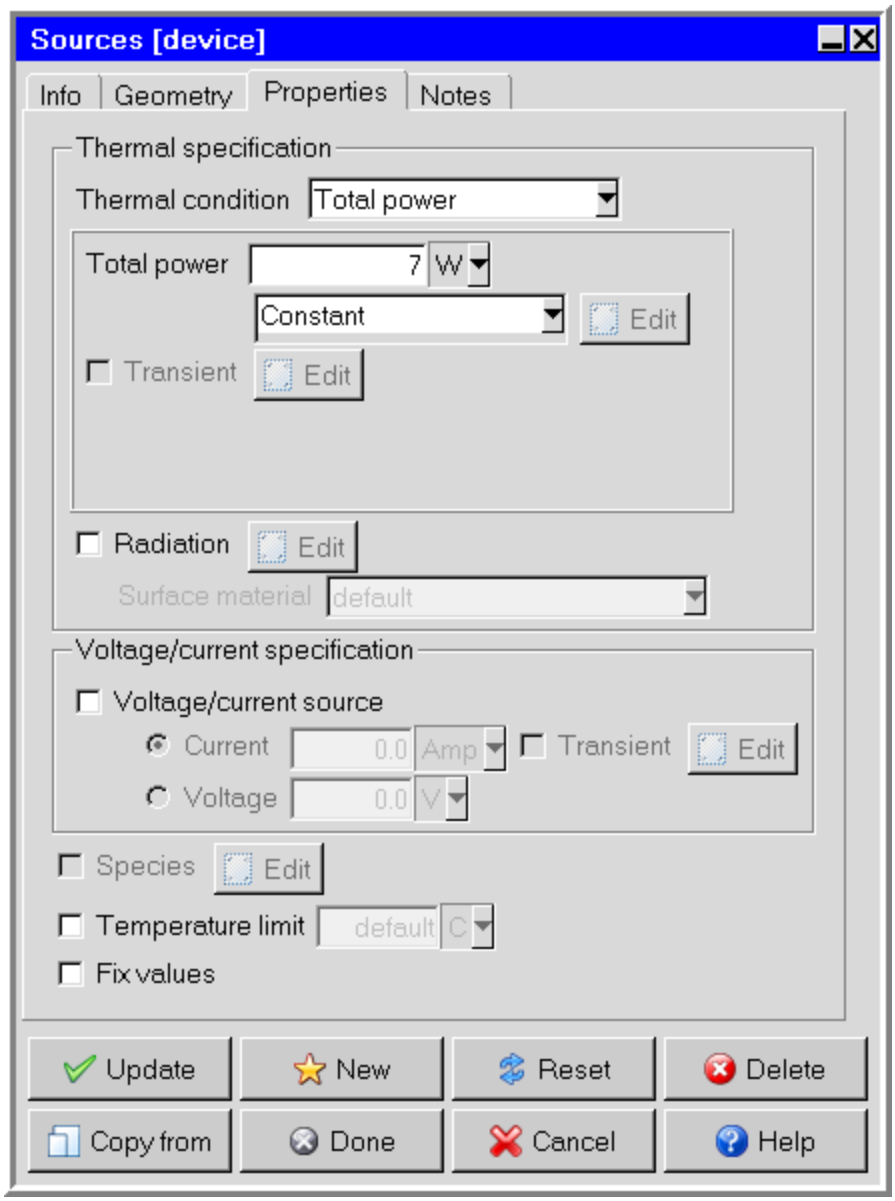
Figure 3.9: The Sources Panel



d. Create two copies of the source and separate them by 0.055 m in the Z direction. Follow the steps below for copying the source object.

   i. Right-click the source object and choose the **Copy** option. Alternatively, click the source object in the **Model manager** window and press **Ctrl**+**C** to bring up the **Copy** panel.

   ii. Specify the **Number of copies** as **2**.

   iii. Select the **Translate** option.

   iv. Specify a **Z offset** of **`0.055`** m.

   v. Click **Apply** to copy the object.

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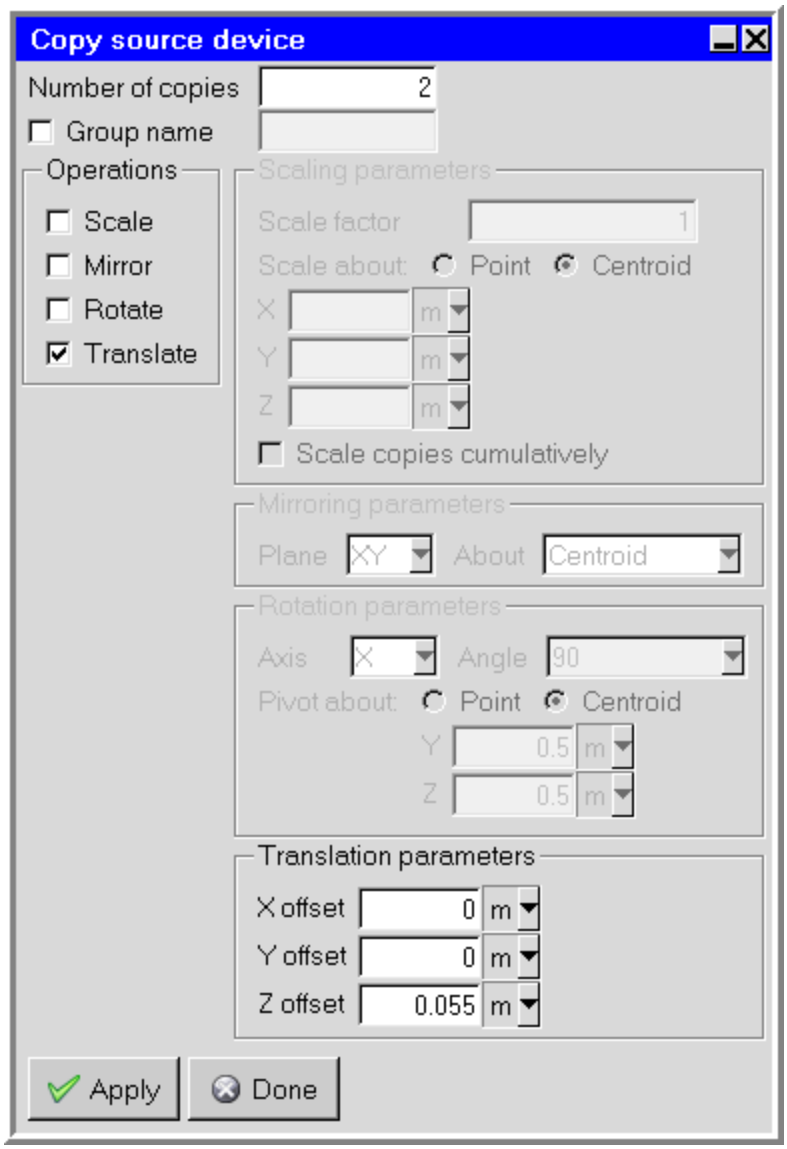
Figure 3.10: The Copy source device Panel



e. Similarly, create the other source objects (devices) by copying the sources created in the previous steps.

   i. Click and select `device`, then while holding down the **Ctrl** key, select `device.1`, and `device.2`. Right-click and choose the **Copy** option, or press **Ctrl**+**c**.

   ii. Specify the **Number of copies** as **3**.

   iii. Turn on the **Translate** option.

   iv. Specify a **Y offset** of **`0.064`** m.

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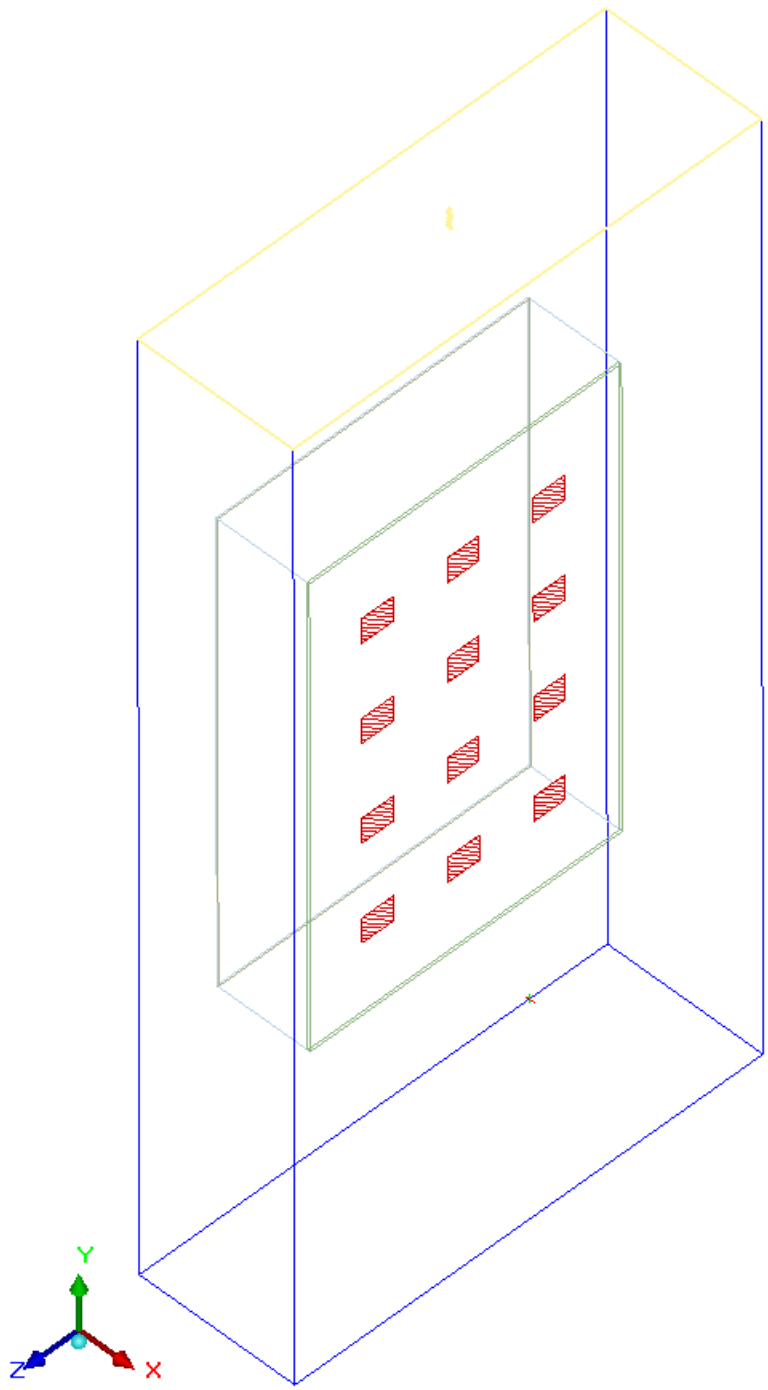
v. Click **Apply** to copy the object.

Note:

Following these two copy actions, you should now have 12 sources (Figure 3.11: Geometry with Devices (p. 64)) in a four rows by three columns pattern.

Figure 3.11: Geometry with Devices



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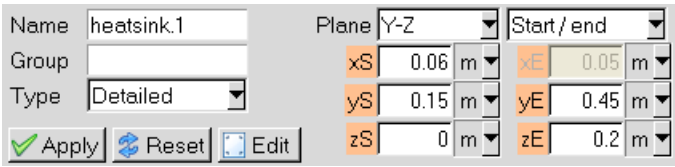
7. Create the heat sink.

   You will create an extruded fin heat sink with the flow in the Y direction to remove heat from the PCB.

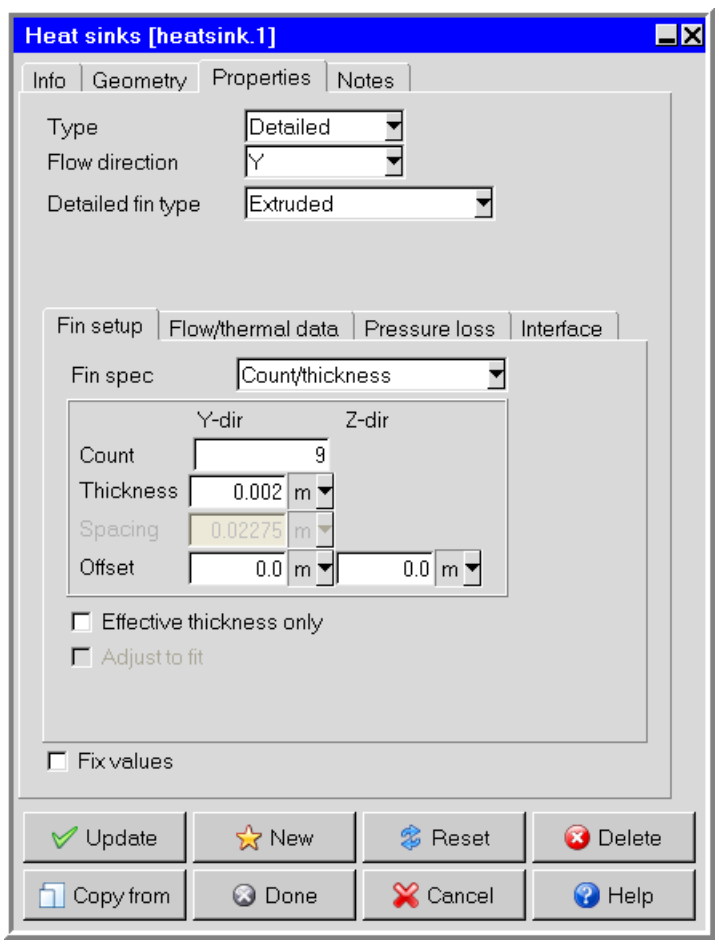
   a. Click the **Create heat sinks** icon ( ) in the **Model** toolbar to create a heat sink and double-click the heat sink object in the **Model manager** window. Specify the following dimensions in the geometry window.



   b. In the **Heat sinks** object panel, select the **Geometry** tab, and specify a **Base height** of **`0.004`** m and an **Overall height** of **`0.04`** m.

   c. Specify the properties of the heat sink as shown in Figure 3.12: The Heat sinks Panel (p. 65) below. Note that you are not changing parameters in the **Flow/thermal data**, **Pressure loss**, or **Interface** tabs.

   Figure 3.12: The Heat sinks Panel



   d. Click **Done** to close the **Heat sinks** panel.

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8. Create the fan.

   For this model, we will make use of Ansys Icepak's fan library and search tool.

   a. Select the **Library** tab in the model manager window (Figure 3.13: Search Fan library Panel (p. 67)).

   b. Right-click **Libraries** in the **Model manager** window and choose **Search fans**.

      The **Search fan library** dialog appears.

      i. In the **Physical** tab, deactivate the **Min fan size** and enter **`80`** mm for the **Max fan size**.

      ii. Select the **Thermal/flow** tab, enable the **Min flow rate** option and specify a **Min flow rate** of **`80`** cfm.

      > **Note:**
      >
      > The minimum flow rate used in the search criteria implies the minimum free flow of the fans.

      iii. Click the **Search** button.

      > **Note:**
      >
      > Ansys Icepak lists all the fans in its libraries that satisfy these conditions.

   c. Select the fan called `delta.FFB0812_24EHE` in the **Name** column by clicking it.

   d. Click **Create** to load the fan into the model.

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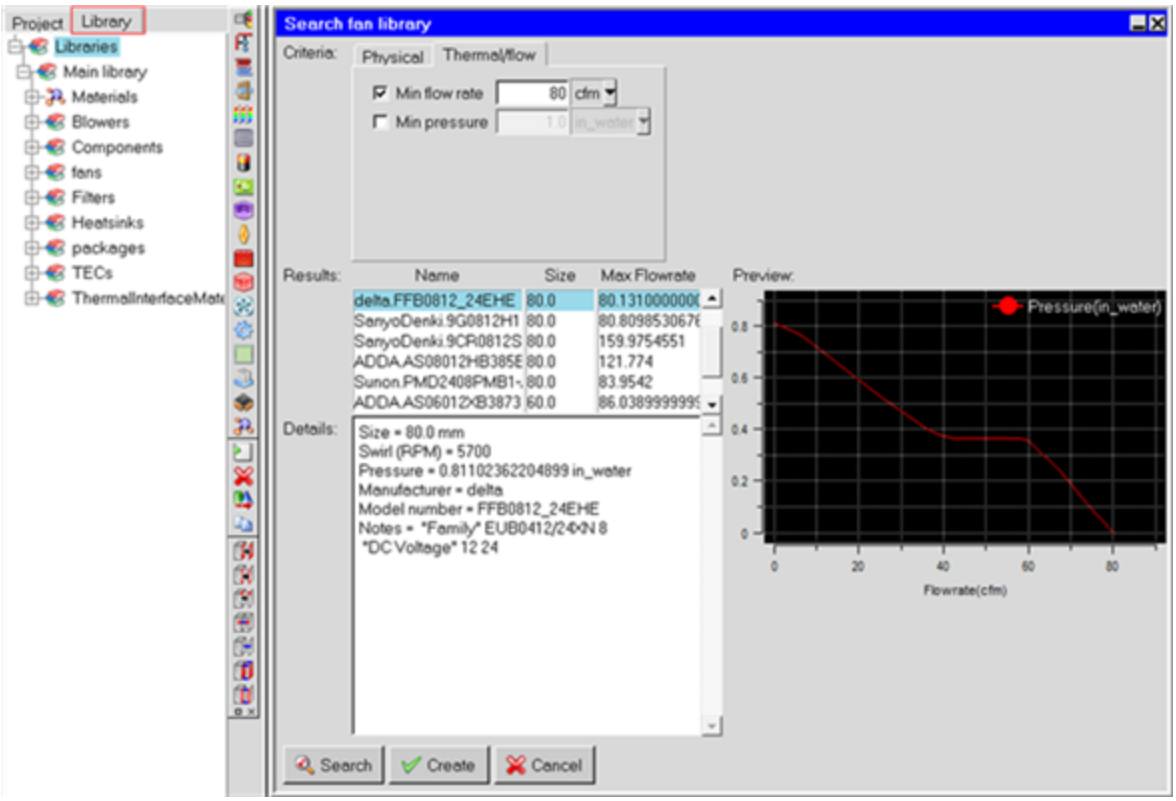
Figure 3.13: Search Fan library Panel



e. Now, specify the location of the fan. Resize the fan geometry as shown in Figure 3.14: The Fans Panel (p. 68). Note the plane orientation is **X-Z**.

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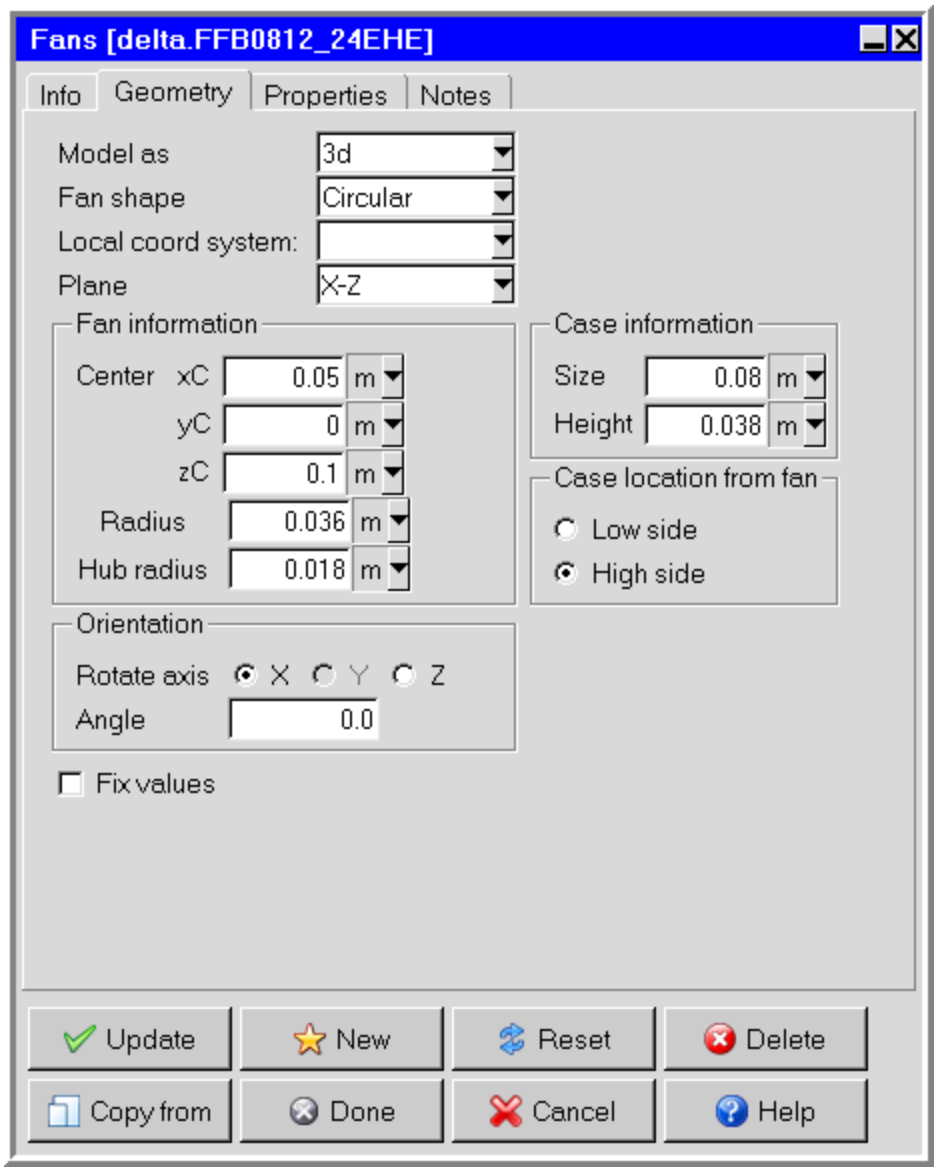
Figure 3.14: The Fans Panel



The final geometry should look like Figure 3.15: The Final Geometry (p. 69).

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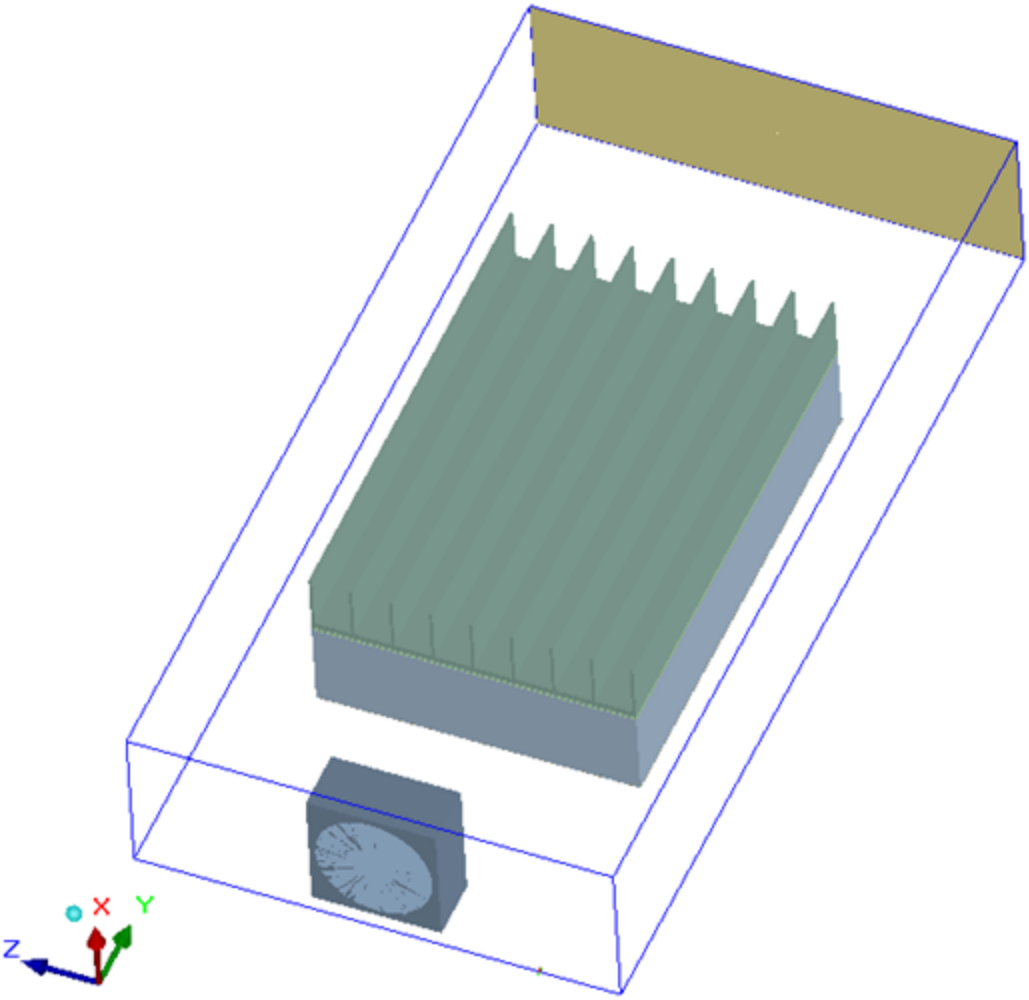
Figure 3.15: The Final Geometry



f. Display objects by type.

You can enable the display of all object types, the filtering of blocks by type (fluid, solid, network, hollow), and the display of blocks with traces and/or CAD block. This feature is useful here to view the otherwise unseen devices behind the heat sink. Generally, this also aids in model verification.

i. **Model** → **Show objects by type**

ii. The **Show objects by type** panel appears as shown in Figure 3.16: Show objects by type Panel (p. 70).

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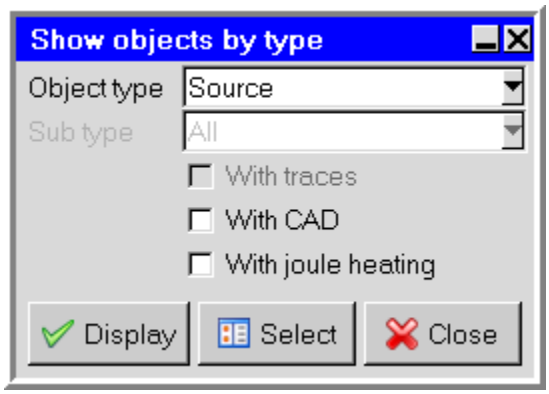
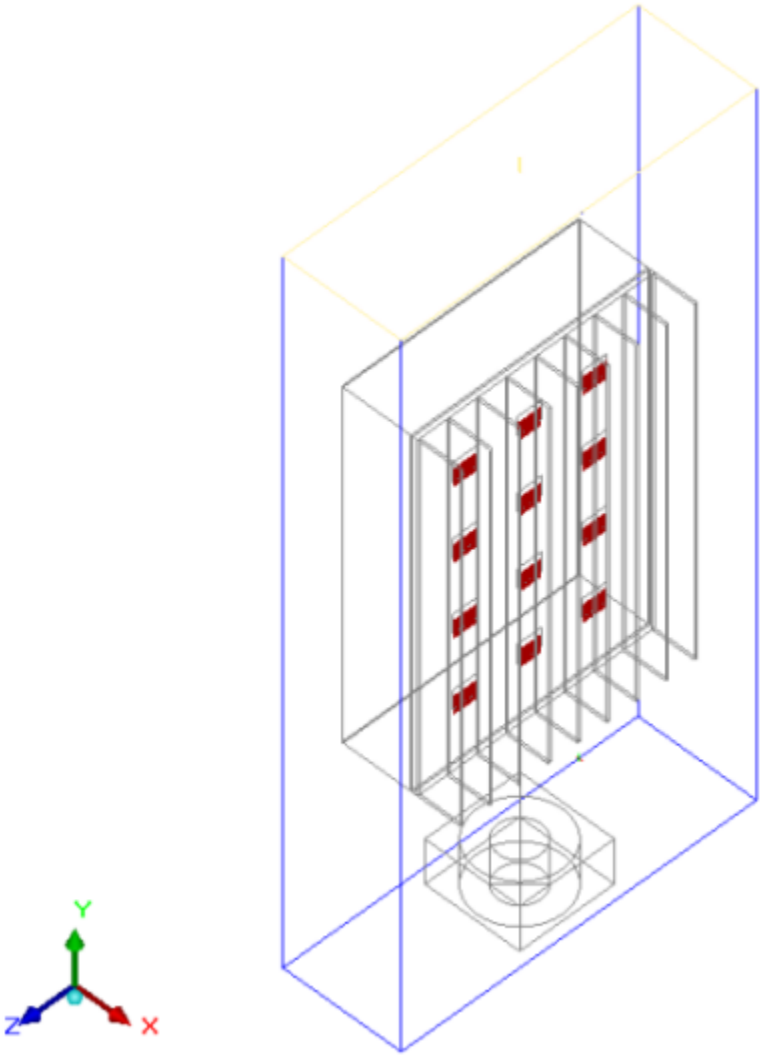
Figure 3.16: Show objects by type Panel



iii. Click **Display** to show the devices. The model then appears as in Figure 3.17: Displaying Objects of Type Source (p. 70).

Figure 3.17: Displaying Objects of Type Source



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iv. Click **Close** to exit the **Show objects by type** panel.

g. Check the definition of the modeling objects to ensure that you specified them properly.

View → **Summary (HTML)**

Note:

The HTML version of the summary displays in your web browser. The summary displays a list of all the objects in the model and all the parameters that have been set for each object. You can view the detailed version of the summary by clicking the appropriate object names or property specifications. If you notice any incorrect specifications, you can return to the appropriate modeling object panel and change the settings in the same way that you originally entered them.

3.6. Step 3: Create Assemblies

For both organizational purposes and to have a finer mesh in the fan and enclosure, you will create two assemblies. The first assembly consists of the RF amplifier and heat sink; the second assembly consists only of the fan.

1. To create the amplifier assembly:

   a. Select the **positive X** view by either using the icon in the shortcut menu or simply press **Shift**+**X** and then **S** to scale to fit the view in the graphics window.

   b. While pressing **Shift**, drag a bounding box around the amplifier using the left mouse button. Do not drag the bounding box around the fan. Release the mouse button and notice that all of the objects forming the amplifier and heat sink have been selected in the **Model manager** window.

   c. Right-click the highlighted enclosure ("Housing") in the **Model manager** window and select **Create** and then **Assembly** from the list. You have now added all of the selected objects to the assembly.

   d. In the **Object geometry window**, rename `assembly.1` to `amplifier` and click **Apply**.

2. Create a new assembly for the fan object:

   a. Click the **Create assemblies** icon ( ) in the model toolbar to create a new assembly.

   b. In the **Model manager** window, use the left mouse button to drag the fan, `delta.FFB0812_24EHE`, into the new assembly to add the fan to this assembly.

   c. In the **Object geometry window**, rename this assembly as `fan` and click **Apply**.

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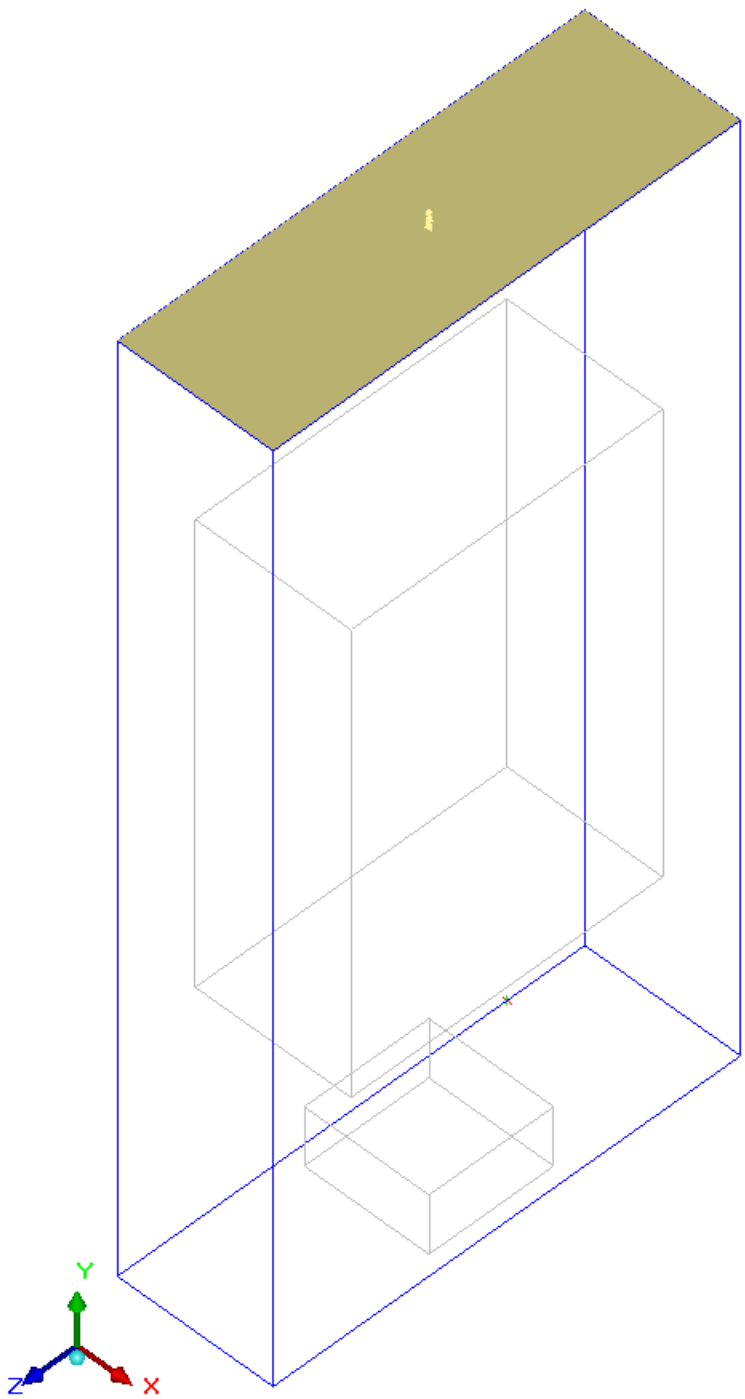
Figure 3.18: Two Assemblies



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3.7. Step 4: Generate a Mesh

Before generating a mesh, specify the slack values for the assemblies.

Note:

Slack values represent a finite offset from an object to a **non-conformal** mesh boundary and are required when meshing assemblies separately. Selecting small slack values can decrease the total number of cells in the mesh with a negligible change in accuracy. On the other hand slack values that are too large may cause excessive mesh bleeding. It is good practice to set slack values such that two or three cells fit in the slack region. Note that in this particular model the gap between the two assemblies is large enough to accommodate nonzero slack values. Refer to Zero Slack with Non-Conformal Meshing (p. 435) for more discussion on slack values.

1. Edit both assemblies (right-click the assembly name in the model toolbar and select **Edit**), then select the **Meshing** tab.

2. Toggle **Mesh separately** and then specify the slack values indicated in the following table. Make sure you remember to add slack values to *both* assemblies.

Table 3.1: Slack Values for the Amplifier and Fan

Name	Min X	Min Y	Min Z	Max X	Max Y	Max Z
Amplifier	0	0.02	0.01	0	0.05	0.01
Fan	0.01	0	0.01	0.01	0.05	0.01

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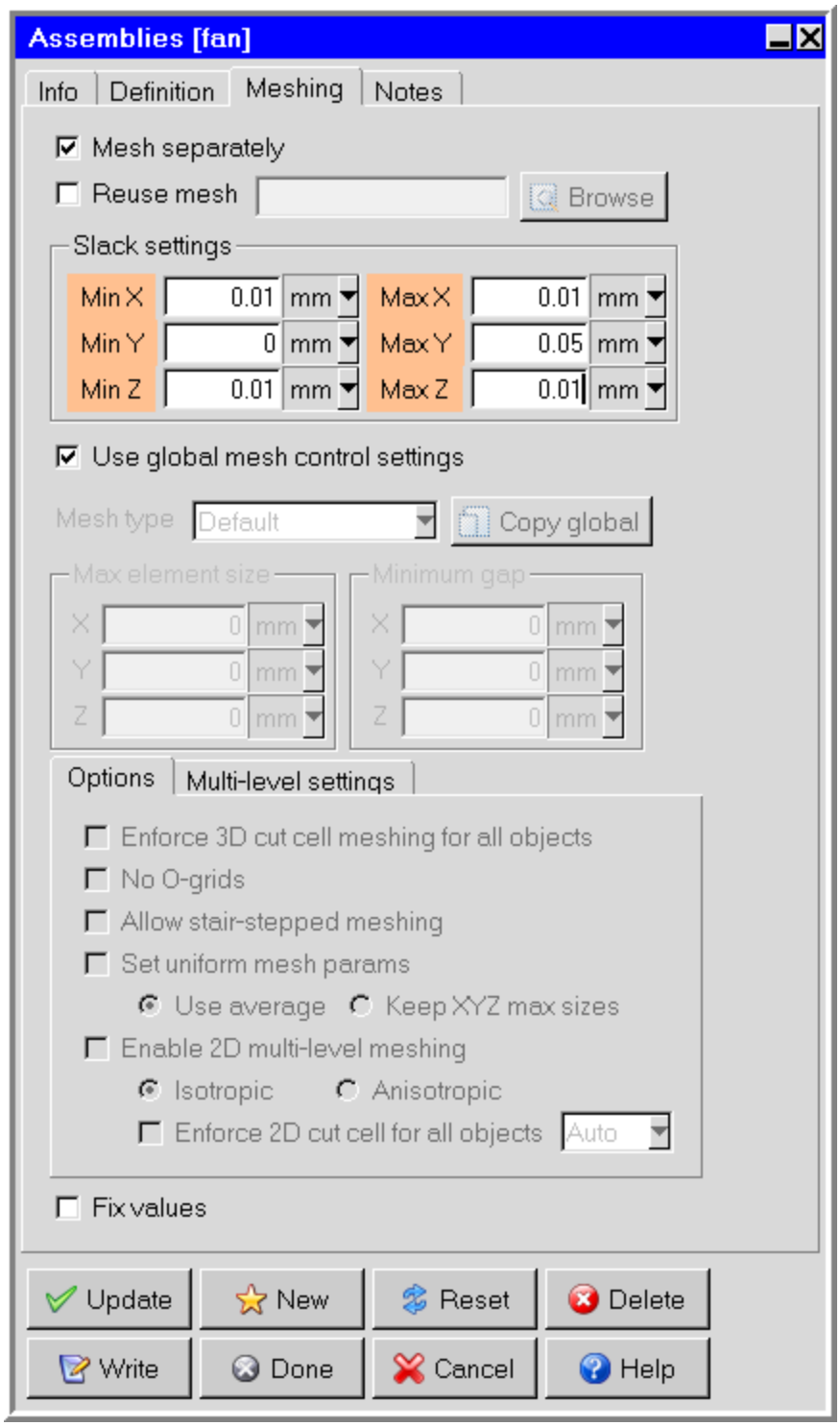
Figure 3.19: Fan Assemblies Panel



3. To create the mesh, go to **Model** → **Generate Mesh**. The **Mesh control** panel (Figure 3.20: The Mesh control Panel (p. 75)) appears. You can also open the **Mesh control** panel by clicking the **Generate mesh** icon ( ) in the shortcut menu.

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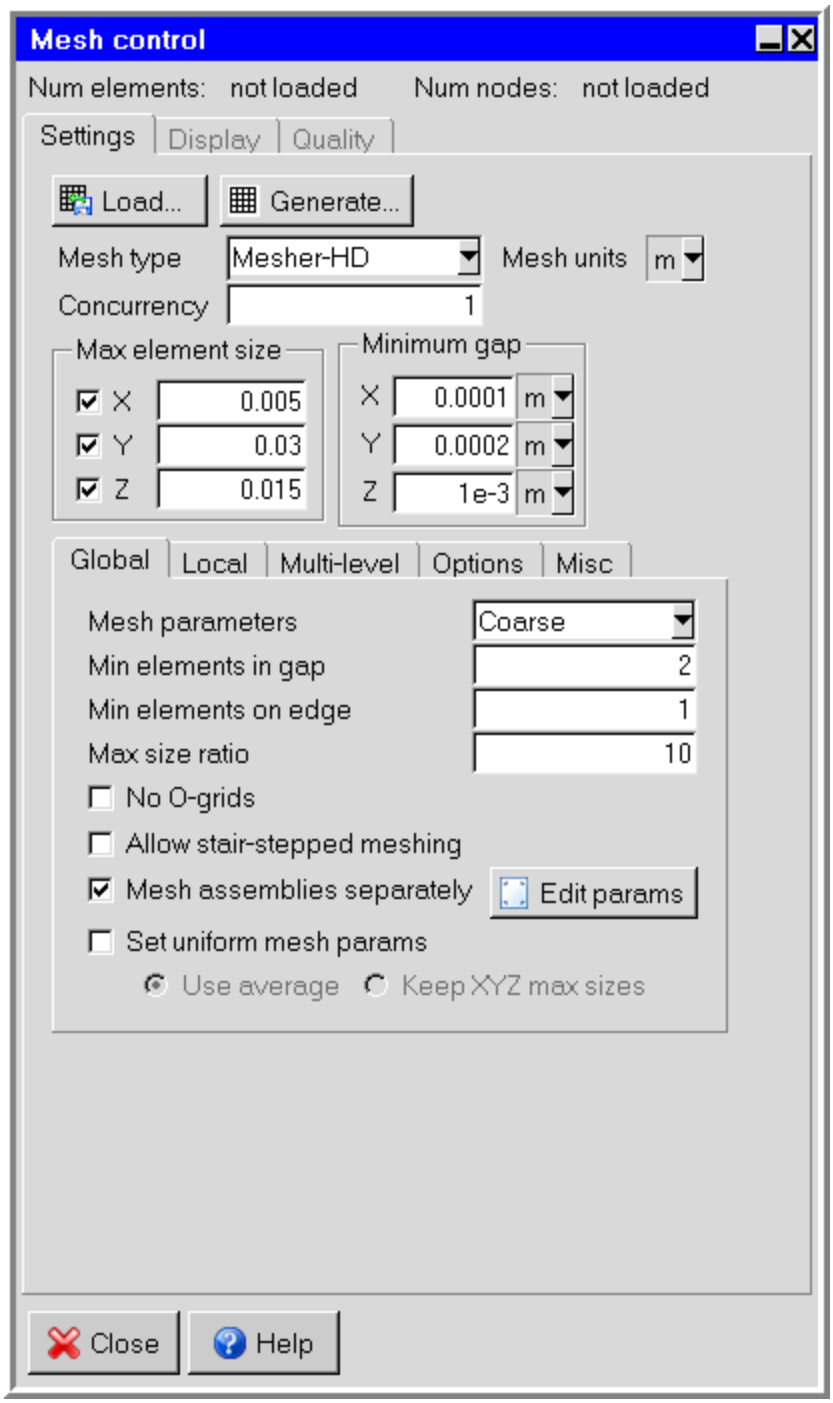
Figure 3.20: The Mesh control Panel



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4. First, generate a coarse mesh by choosing **Coarse** in the **Mesh parameters** drop-down list in the **Global** tab, as shown in Figure 3.20: The Mesh control Panel (p. 75). Click **Generate** to create a mesh.

   Note:

   If you have unchecked **Allow minimum gap changes** in the **Misc** tab, the **Minimum separation** warning will appear. This warning message appears when the minimum gap specified is more than 10% of the smallest sized object in the model. Select **Change value and mesh** if the warning message pops up.

5. To view the mesh, display a plane cut view through the center of the cabinet, perpendicular to the fins (y-z plane).

6. To create a plane cut, follow these steps:

   a. Click the **Display** tab at the top of the **Mesh control** panel.

   b. Toggle **Display mesh** and **Cut plane**.

   c. Under **Plane location**, set position to **X plane through center** in the drop-down list.

   d. Press **Shift**+**X** to orient to the **positive X** direction and view the newly created cut plane.

   e. Move the plane using the slider bar to see different views.

   Make sure that the amplifier assembly is expanded and inspect the cells adjacent to the heat sink fins. Notice that the resolution is coarse (Figure 3.21: Coarse and Fine Mesh (p. 77)), with only a couple of cells between the fins. As the flow passes between the fins, the boundary layers will grow and their degree of resolution will dictate the accuracy of the simulation. You want to have at least three to four cells between the fins to adequately resolve the growth of the boundary layers. Refining the mesh results in better resolution.

7. Select **Normal** in the **Mesh parameters** drop-down list in the **Settings** tab. Click **Generate** and inspect the resulting mesh. Note that this has increased the number of cells between adjacent fins (Figure 3.21: Coarse and Fine Mesh (p. 77)), providing better resolution of the boundary layers.

   You can display the mesh on selected objects or the cut plane by using the context menu in the graphics window. To display the context menu, hold down the **Shift** key and press the right mouse button anywhere in the graphics window. Select **Display mesh** or **Display cut plane mesh** in the context menu and the mesh will be displayed on selected objects or the cut plane will be displayed.

   It is also a good practice to select the **Quality** tab and review the **Face Alignment**, **Volume**, and **Skewness**. The histograms show the quality metric (Face Alignment, Volume or Skewness) versus the number of cells. Clicking the bars that form the histogram displays the cells with that value in the graphics window.

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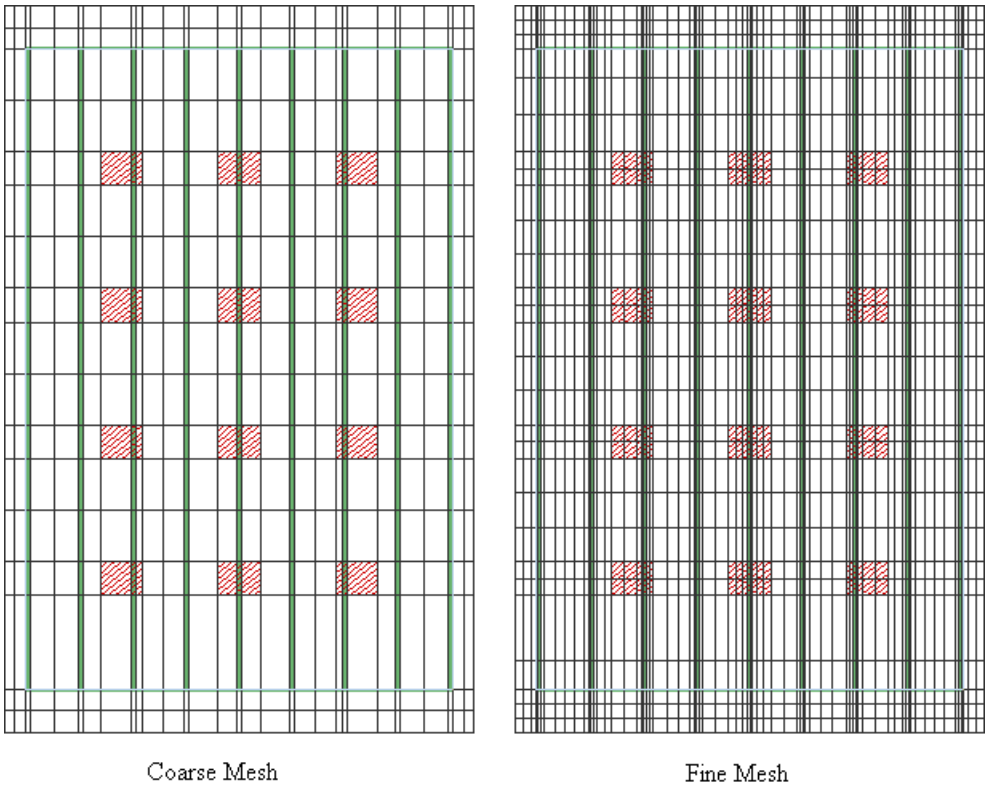
Figure 3.21: Coarse and Fine Mesh



8. Once you have explored the mesh quality, click **Close** to dismiss the **Mesh control** dialog box.

3.8. Step 5: Physical and Numerical Settings

Before starting the solver, you will first review estimates of the Reynolds and Peclet numbers to check that the proper flow regime is being modeled.

1. Check the values of the Reynolds and Peclet numbers.

   Solution settings → **Basic settings**

   a. Click the **Reset** button (Figure 3.22: The Basic settings Panel (p. 78)).

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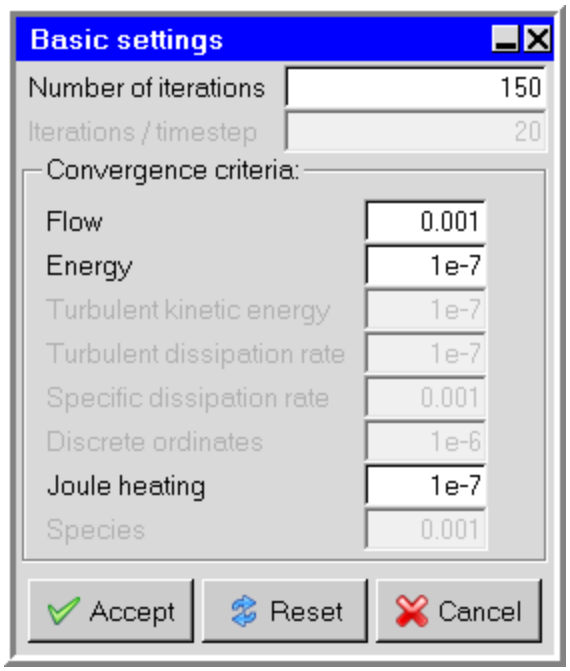
Figure 3.22: The Basic settings Panel



b. Check the values printed to the **Message** window.

The Reynolds and Peclet numbers are approximately 55000.0 and 40000.0 respectively, so the flow is turbulent. Ansys Icepak recommends setting the flow regime to turbulent.

Note:

These values are only estimates, based on the current model setup. Actual values may vary, and may need to be verified, depending on your design.

---

c. Click **Accept** to save the solver settings.

2. Using the **Problem setup wizard**, enable turbulence modeling using the zero equation turbulence model and neglecting radiation heat transfer.

a. In the **Model manager** window, right-click **Problem setup** ( ) and then select **Problem setup wizard**. Figure 3.23: Problem setup wizard panel at step 1 of 14 (p. 79) shows what appears. The **Problem setup wizard** provides a simple interface with user guidance for defining the physics of the model.

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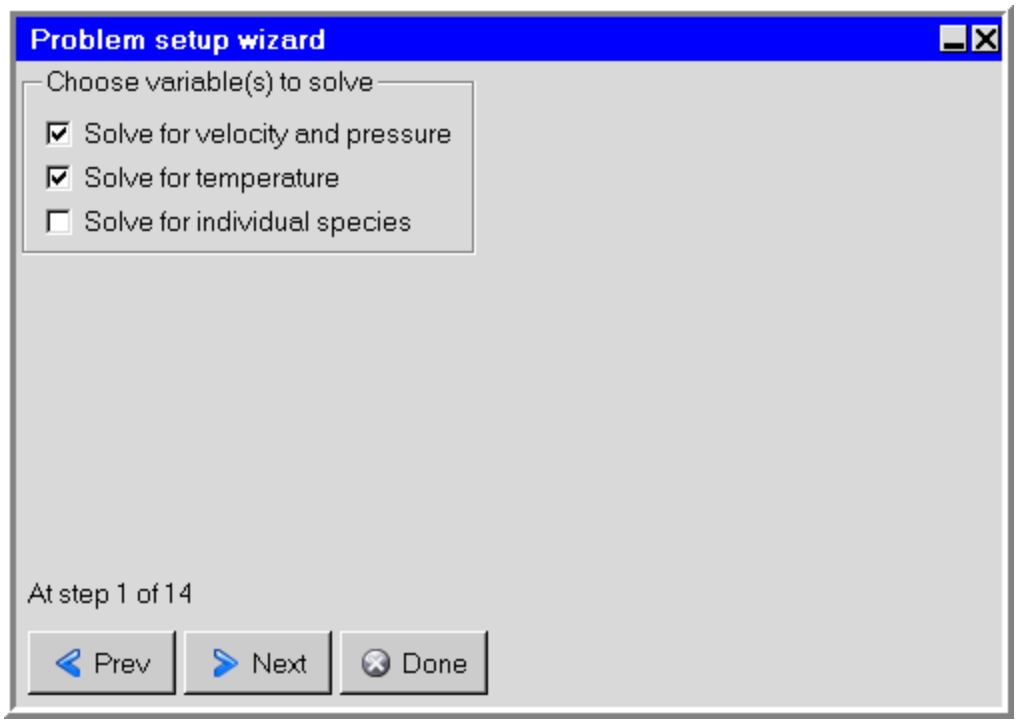
Figure 3.23: Problem setup wizard panel at step 1 of 14



b. For step 1 of 14, indicated in the bottom-left corner of the **Problem setup wizard** panel, keep the default settings for the check boxes. Click **Next**.

c. For step 2 of 14, select **Flow is buoyancy driven (natural convection)** as in Figure 3.24: Problem setup wizard panel at step 2 of 14 (p. 80), since natural convection is not negligible inside the RF amplifier. Click **Next**.

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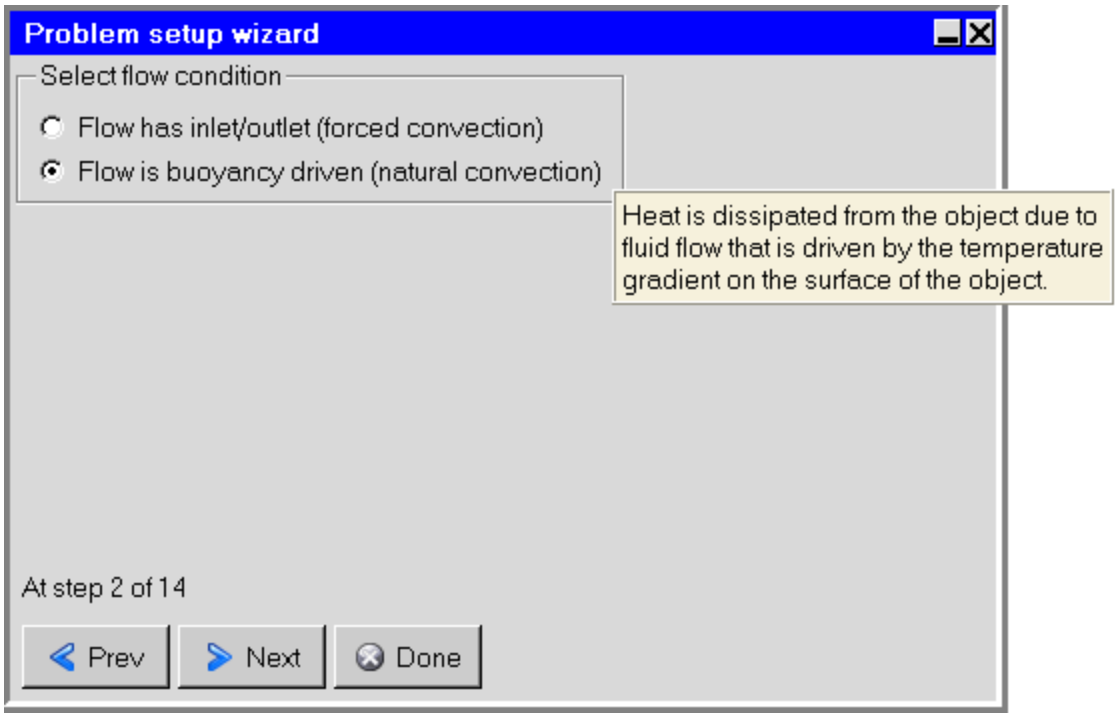
Figure 3.24: Problem setup wizard panel at step 2 of 14



Note:

Hold your mouse pointer over any selection in the **Problem setup wizard** to have a text bubble appear for additional information on the selection, as shown in Figure 3.24: Problem setup wizard panel at step 2 of 14 (p. 80).

d. For step 3 of 14, keep the default setting of **Use Boussinesq approximation** for the natural convection model.

e. For step 4 of 14, keep the **Operating pressure** at the default value of **`101325.0`** N/m2. Select **Set gravitational acceleration** and leave the default values for all directions, as shown in Figure 3.25: Problem setup wizard at step 4 of 14 (p. 81). Click **Next**.

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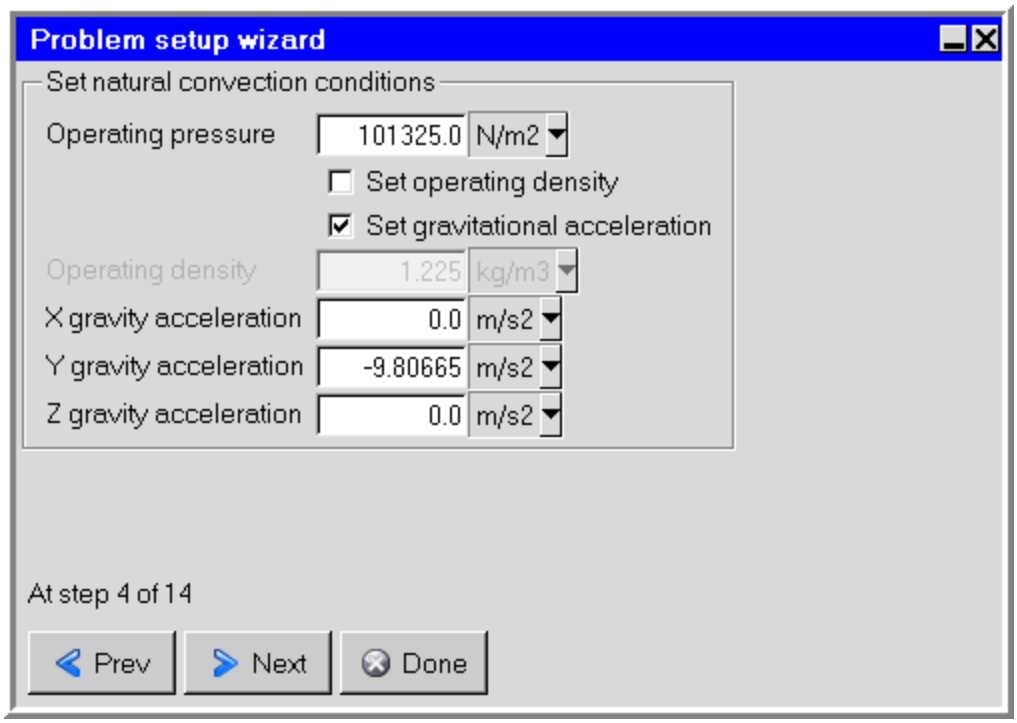
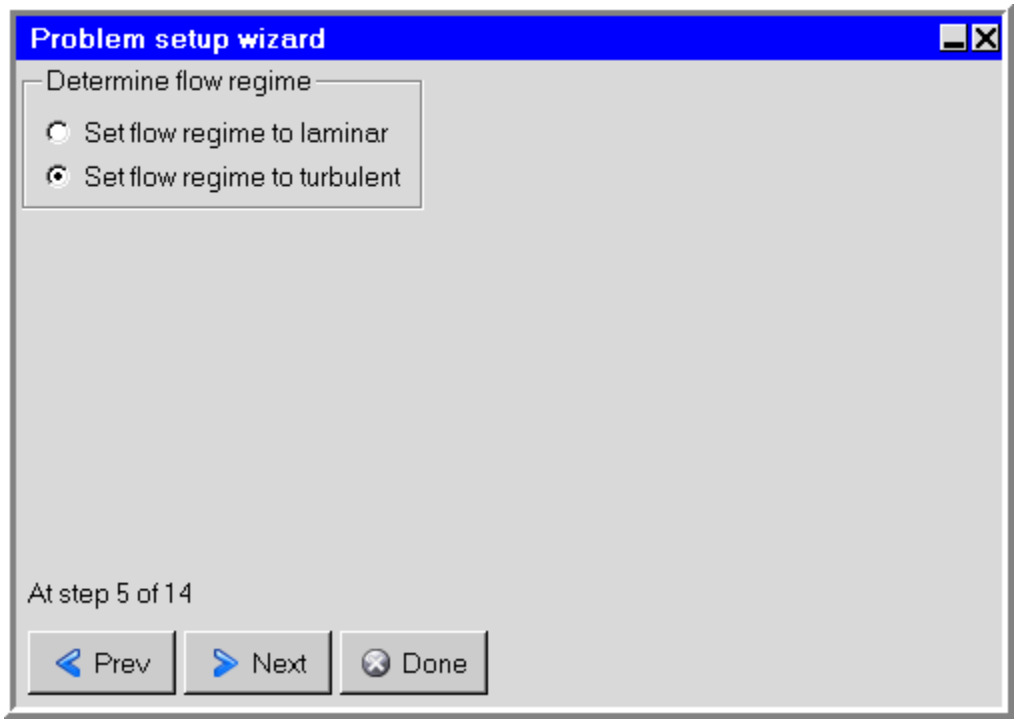
Figure 3.25: Problem setup wizard at step 4 of 14



f. For step 5 of 14, select **Set flow regime to turbulent** to enable turbulence modeling, shown in Figure 3.26: Problem setup wizard at step 5 of 14 (p. 81). Click **Next**.

Figure 3.26: Problem setup wizard at step 5 of 14



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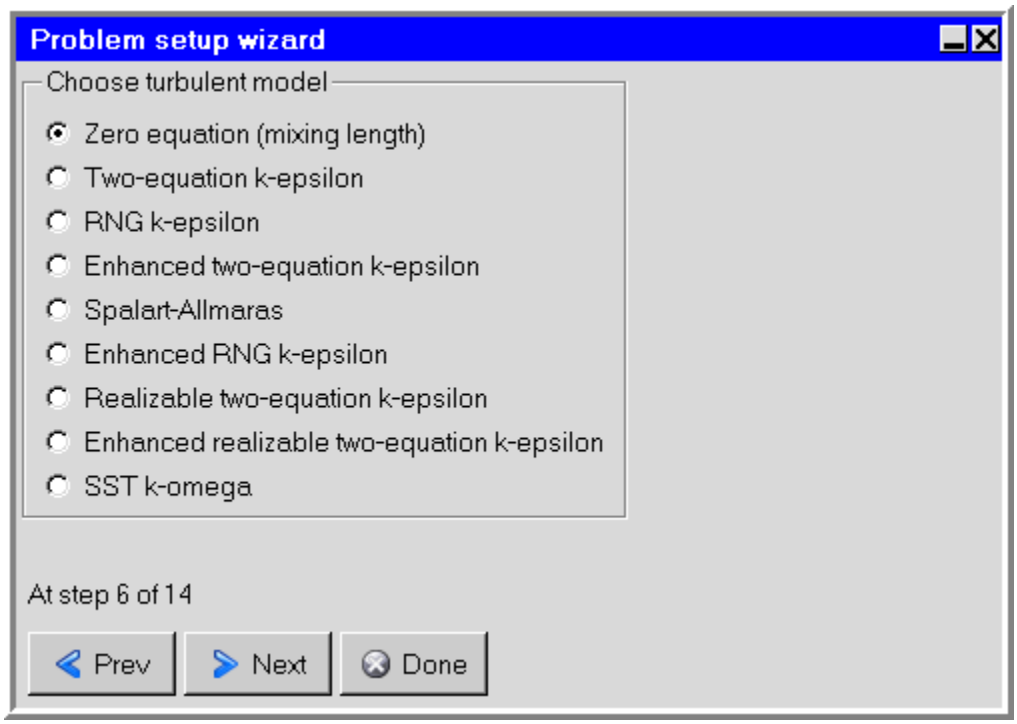
g. For step 6 of 14, select **Zero equation (mixing length)** to choose the zero equation turbulent model, shown in Figure 3.27: Problem setup wizard at step 6 of 14 (p. 82). Click **Next**.

Figure 3.27: Problem setup wizard at step 6 of 14



h. For step 7 of 14, turn off radiation by selecting **Ignore heat transfer due to radiation**. Click **Next**.

i. For step 9 of 14, keep the check box clear to ignore solar radiation. Click **Next**.

j. For step 10 of 14, keep the default setting of **Variables do not vary with time (steady-state)** for steady-state simulation. Click **Next**.

k. For step 14 of 14, keep the check boxes clear to ignore altitude effects. Finally, click **Done** to finish the **Problem setup wizard**. These settings now fully define the problem setup.

Note:

You can edit these settings and other aspects of the problem setup by double-clicking ( ) **Basic parameters** in the **Model manager** window. Figure 3.28: Basic parameters panel (p. 83) shows the **Basic parameters** panel that appears.

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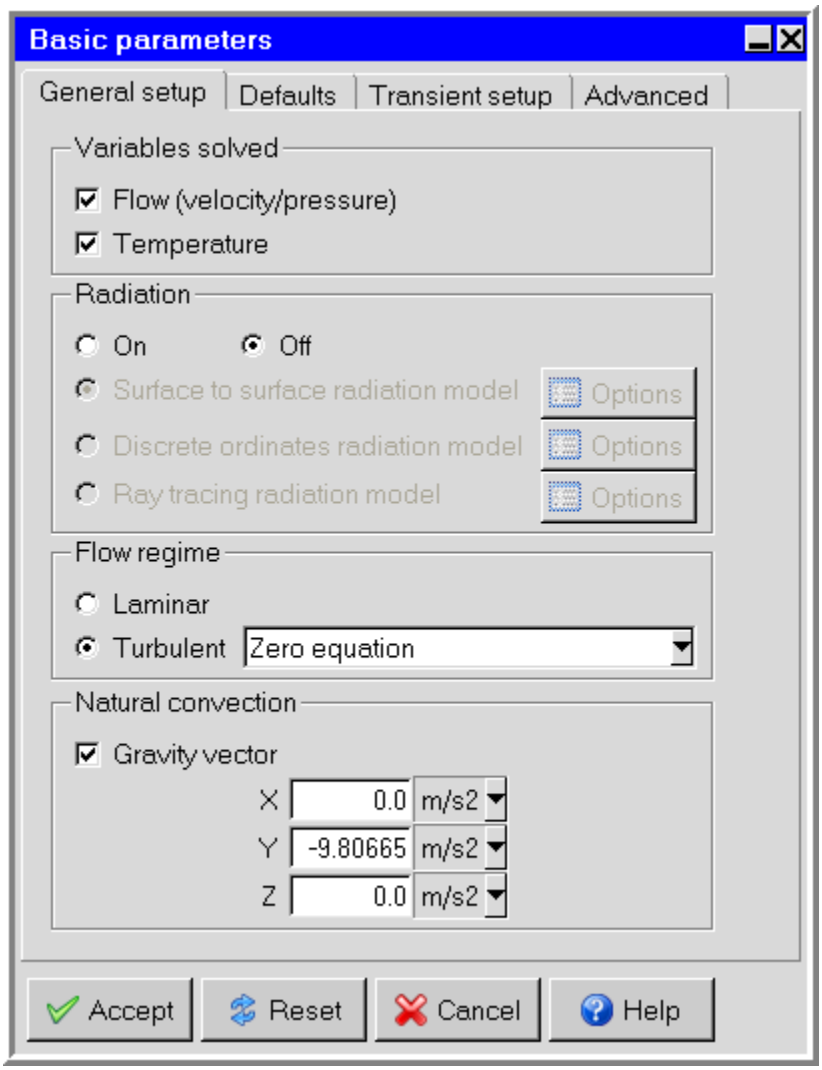
Figure 3.28: Basic parameters panel



3. Return to the **Basic settings** panel, specify the number of iterations as **300**, click **Reset** and then **Accept** again.

4. Set up the temperature limits for all the sources.

   Model → Power and temperature limits

   a. Figure 3.29: Panel of Power and temperature limit setup (p. 84) shows the required settings for the next few steps.

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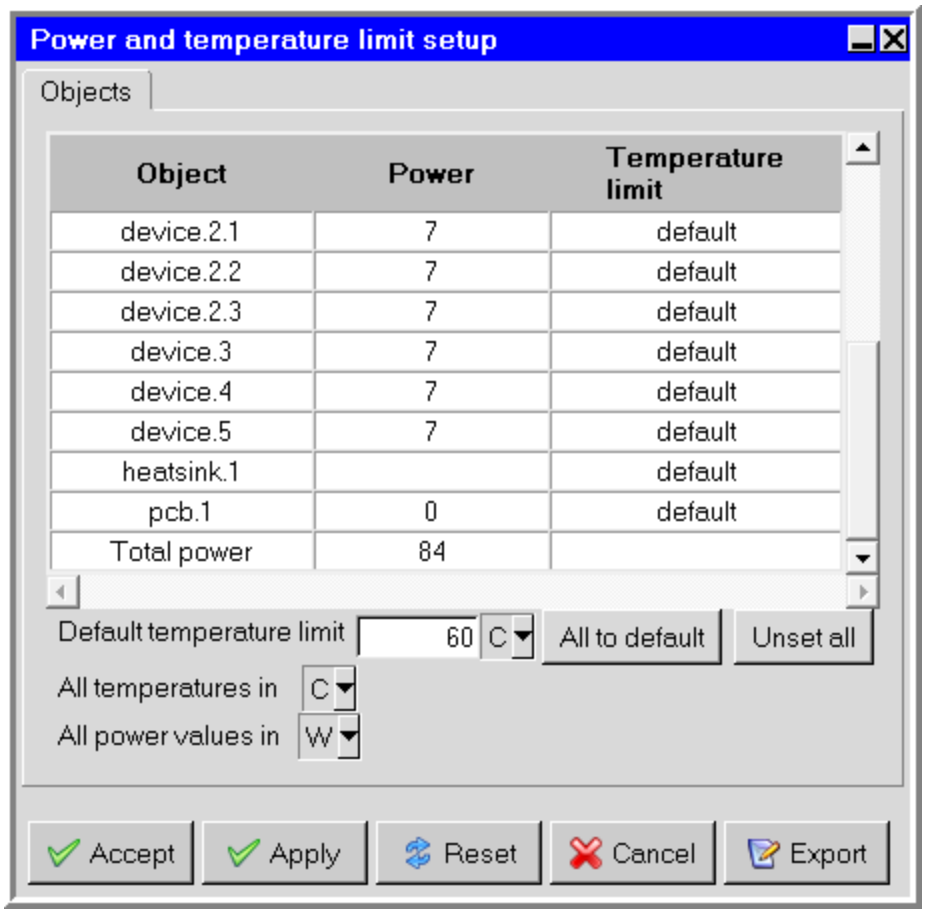
Figure 3.29: Panel of Power and temperature limit setup



b. Enter a new value of **60** C for **Default temperature limit**.

c. Click **All to default**.

d. Click **Apply** and then click **Accept** to close the panel.

Note:

Ansys Icepak uses the default temperature limit during post-processing to identify components that exceed their limits or components that are close to this limit. Ansys Icepak does not use this value to solve the problem.

3.9. Step 6: Save the Model

Ansys Icepak will save the model for you automatically before it starts the calculation, but it is a good idea to save the model yourself as well. If you exit Ansys Icepak before you start the calculation, you will be able to open the job you saved and continue your analysis in a future Ansys Icepak session. (If you start the calculation in the current Ansys Icepak session, Ansys Icepak will simply overwrite your job file when it saves the model.)

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File → Save project

Note:

You can click the save button ( ) in the **File commands** toolbar.

3.10. Step 7: Calculate a Solution

1. Create monitors.

Note:

It is good practice to monitor the solution progress for certain objects. Dragging the object in the **Model manager** window and placing it in the **Points** folder can accomplish this.

---

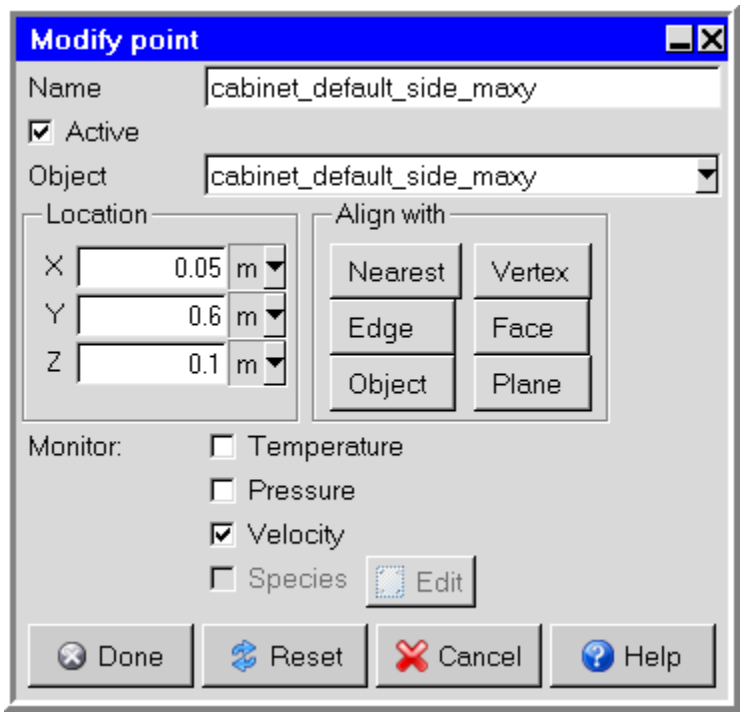
a. Drag `device.2` and `cabinet_default_side_maxY` into the **Points** folder.

b. Right-click the `cabinet_default_side_maxY` in the **Points** folder.

c. Select **Edit** and deselect **`Temperature`** and select **`Velocity`** (Figure 3.30: The Modify point Panel (p. 85)).

d. Click **Done** to accept the modifications and to dismiss the per-object's **Modify point** panel.

Figure 3.30: The Modify point Panel



2. Start the calculation.

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Solve → **Run solution**

a. Since this particular example has neither radiation nor blocks with joule heating, ensure that you select **Disable radiation** and deselect **Disable varying joule heating**.

Note:

Disable radiation instructs Ansys Icepak not to calculate radiation parameters for the model. For cases in which your model has blocks with joule heating, only then should you select **Disable varying joule heating**. Otherwise, leave the option deselected. Refer to Using the Solve Panel to Set the Solver Controls in the Ansys Icepak User's Guide for more information on the settings in the **Solve** panel.

---

b. Select **Write overview of results when finished** in the **Results** tab.

c. Click the **Start solution** button to start the solver. While iterating the solution, windows will appear showing convergence history, Figure 3.31: Convergence Plot (p. 87) and Figure 3.32: Monitor Plot - Temperature (p. 88).

Note:

Alternatively, you can click the **Run solution** icon ( ) in the model and solve toolbar to display the **Run solution** panel.

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Figure 3.31: Convergence Plot

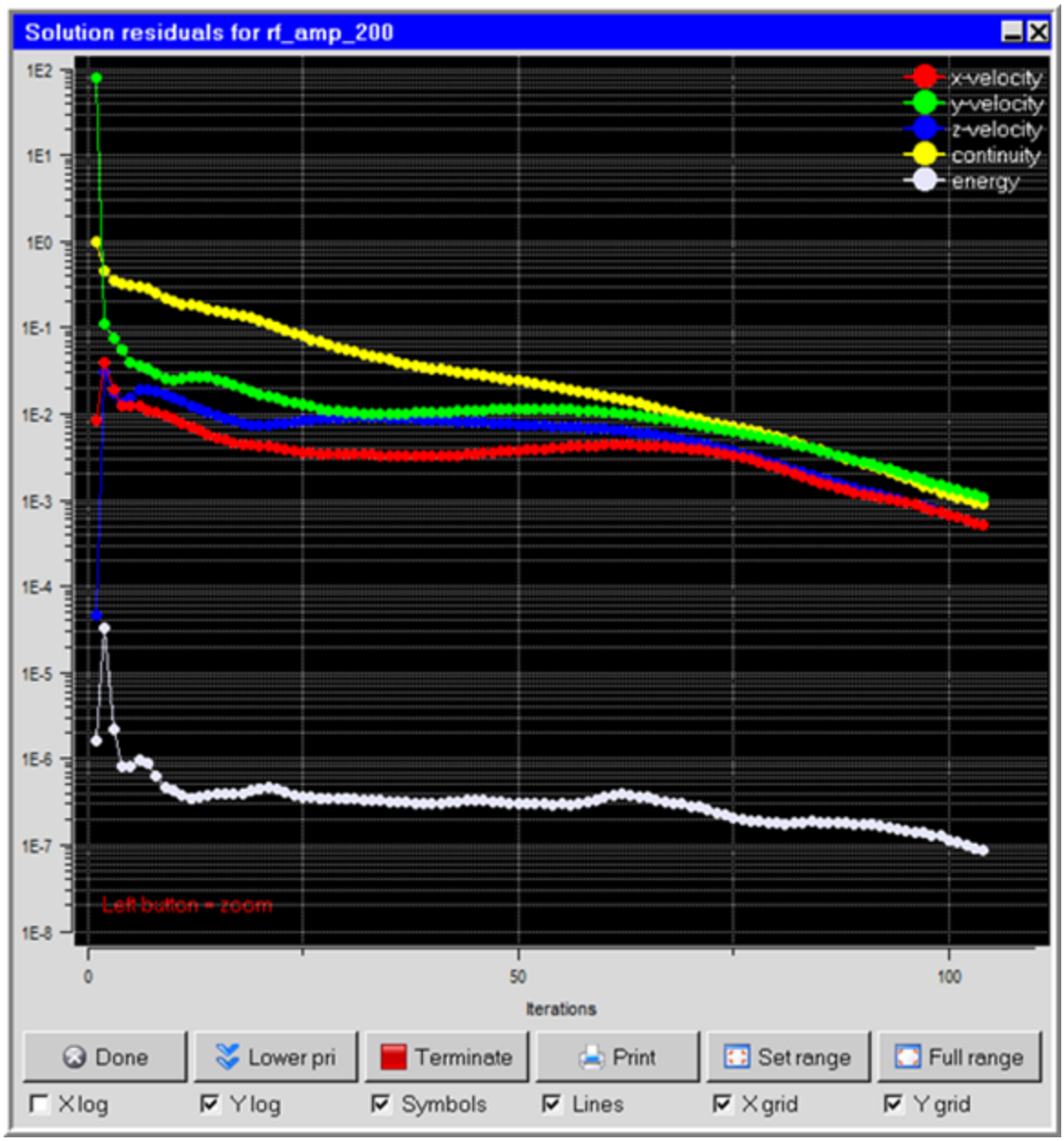


Figure 3.32: Monitor Plot - Temperature

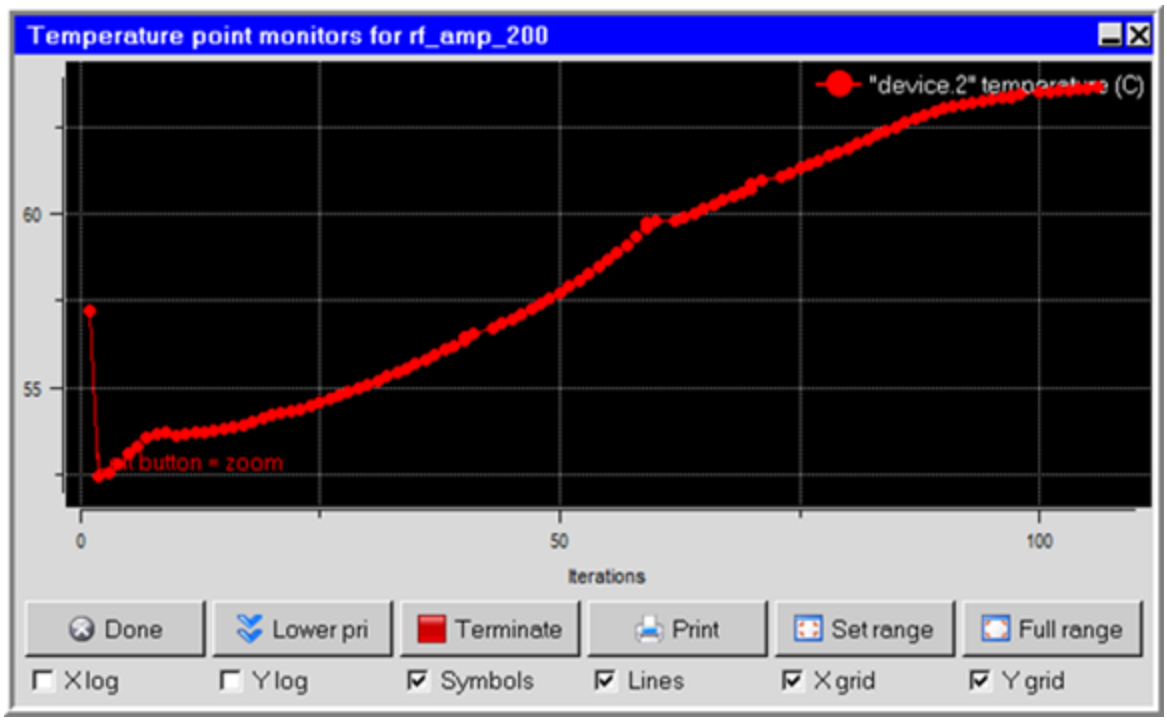
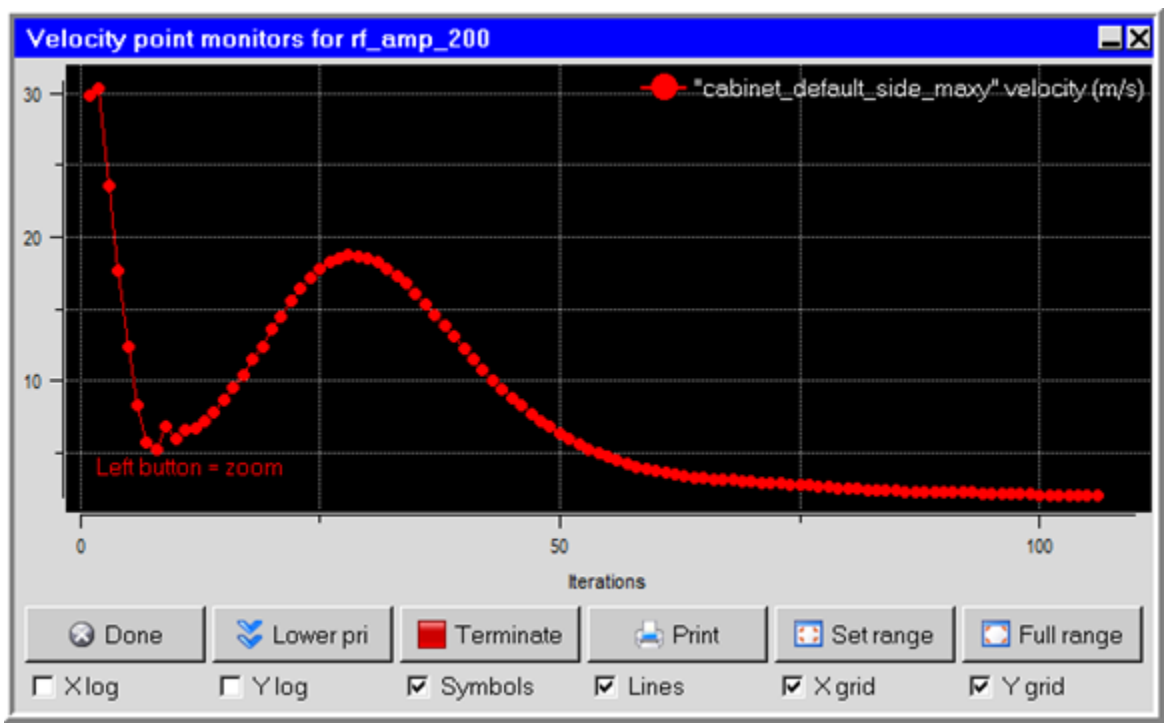


Figure 3.33: Monitor Plot - Velocity



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3.11. Step 8: Examine the Results

Once the model has converged (Figure 3.31: Convergence Plot (p. 87) and Figure 3.32: Monitor Plot - Temperature (p. 88)), Ansys Icepak automatically generates a solution overview report. This report contains detailed information, such as object-based mass and volumetric flow rates, fan operating points, heat flows for objects with specified power, heat flows for objects that communicate with the ambient, maximum temperatures, and overall balances.

Carefully review the solution overview and note that the solution satisfies conservation of mass and energy (scroll to the bottom of the report). Also note the fan operating point. The solution overview is automatically saved and can be reopened from **Report** → **Solution overview** → **Create**.

1. Compare the object temperature values for all sources with the temperature limits assigned.

   Post → **Power and temperature values**

   The **Power and temperature limit setup** panel appears.

   a. Click **Show too hot**.

      The Power and Temperature limit setup show the default temperature limit and the resulting maximum temperature value for each source next to them.

      If an assembly is expanded in the **Model manager** window and if the resulting temperature of any object exceeds the temperature limit specified, Ansys Icepak shows all the critical objects in red color.

   b. Click **Accept** to close the dialog box.

2. Create object faces.

   Note:

   Ensure that the amplifier and fan assemblies are expanded, so that the fins are visible.

   a. Press **Shift**+**Z** to orient the view in the **positive Z** direction.

   b. To create an object face, click the **Object face** icon ( ) in the shortcut toolbar.

   c. In the **Object** drop down list, specify `heatsink.1` as the object and click **Accept**.

   d. Select **Show contours** and click the **Parameters** button (adjacent to **show contours**) to access the **Object face contours** edit dialog box.

      i. Select **This object** in the drop-box adjacent to **Calculated** to use the object-based range.

      ii. Click **Done** to close the **Object face contours** panel.

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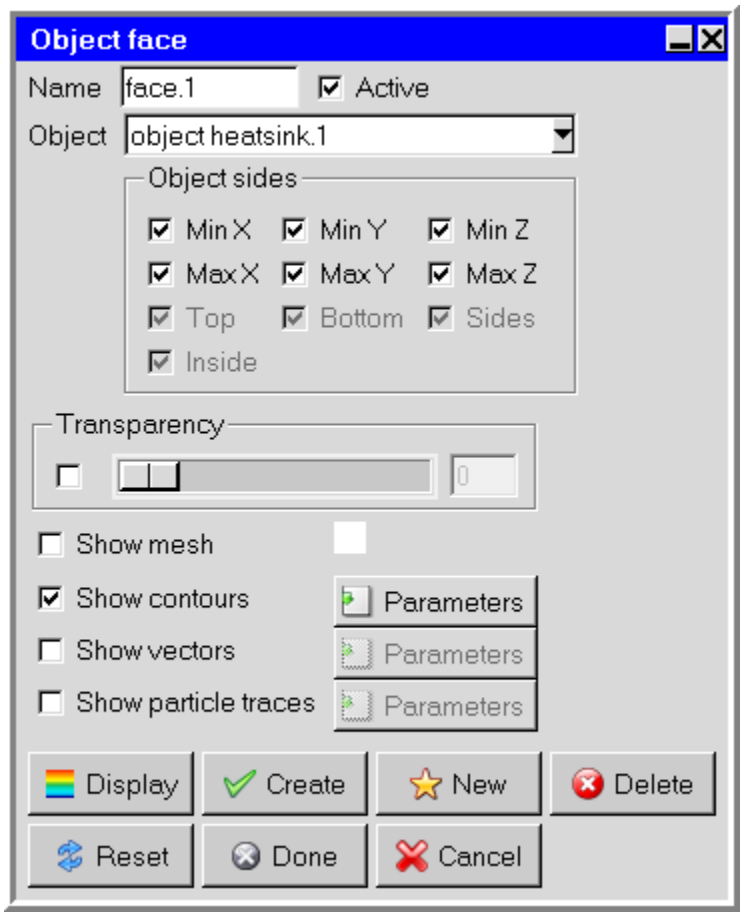
e. Click **Done** to close the **Object face** panel.

Note:

You can also create contours on `heatsink.1` by selecting this object in the **Model manager** window and right-clicking to display the context menu. Select **Create**>**Object face(s)**>**Separate** and the **Object face** panel will appear. The **Object face** panel is displayed for that particular object.

Figure 3.34: Object face Panel



Note:

Using the mouse, rotate the heat sink to examine the surface temperature distribution. Notice that the location of the devices is clearly discernible on the bottom of the heat sink. Also note that the devices get progressively hotter in the flow direction (Figure 3.35: Temperature Contours on the Heat Sink Object Face (p. 91)).

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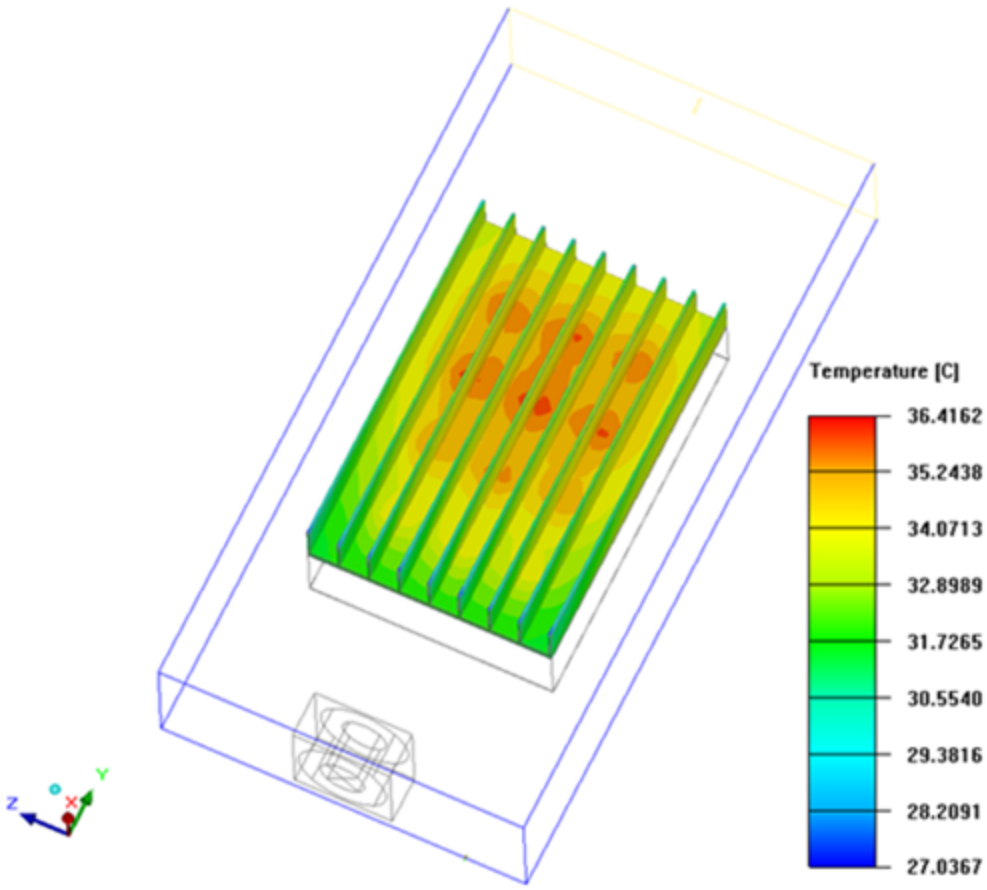
Figure 3.35: Temperature Contours on the Heat Sink Object Face



Note:

Notice that `face.1` has now appeared in the **Model manager** window in the **Post-processing** folder. Right-click `face.1` and note that you can make it active, edit it, or delete it. You can move `face.1` into the **Inactive** folder by dragging and dropping `face.1` within the **Model manager** window. You can make `face.1` active again or delete it by dragging it to **Trash** or to the **Post-processing** folder, as well as with the right-click dialog.

3. Create plane cuts.

   a. To create a plane cut, click the **Plane cut** icon ( ) in the shortcut toolbar.

   b. Select the **Set position** as **Point and normal** and select **Show vectors**, as shown in the panel below. Enter **PX**, **PY**, and **PZ**, as well as **NX**, **NY**, and **NZ** according to Figure 3.36: Plane cut Panel (p. 92). You may also specify the point's coordinates as **`0.680`** by using the slider at the bottom of the **Plane location** box in the same panel.

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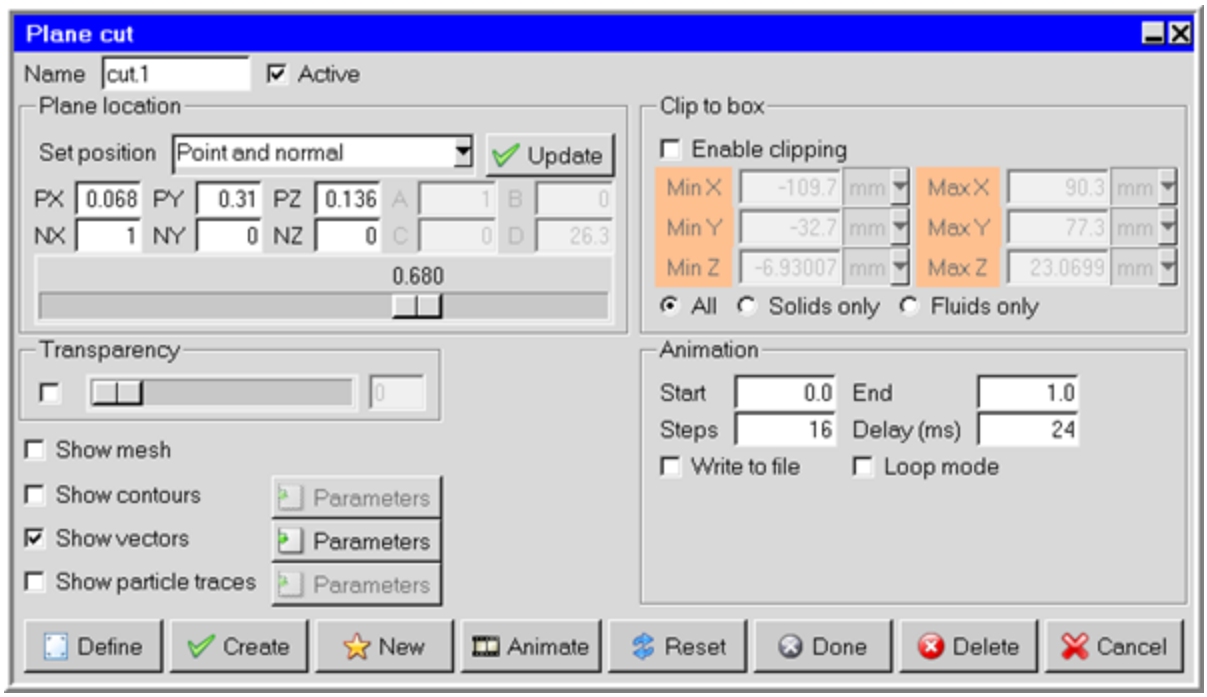
Figure 3.36: Plane cut Panel



c. Click the **Parameters** button adjacent to **Show vectors**.

d. Select **Uniform** in **Display options** group box and specify value as **`5000`**. The **Uniform** option for the velocity will uniformly place the vectors among the 5000 data points.

e. Select **This object** in the drop-box adjacent to **Calculated** and click **Done** to close the panel.

The vector plots are shown in the graphics window (Figure 3.37: Plane Cut of Velocity Vector Field (p. 93)).

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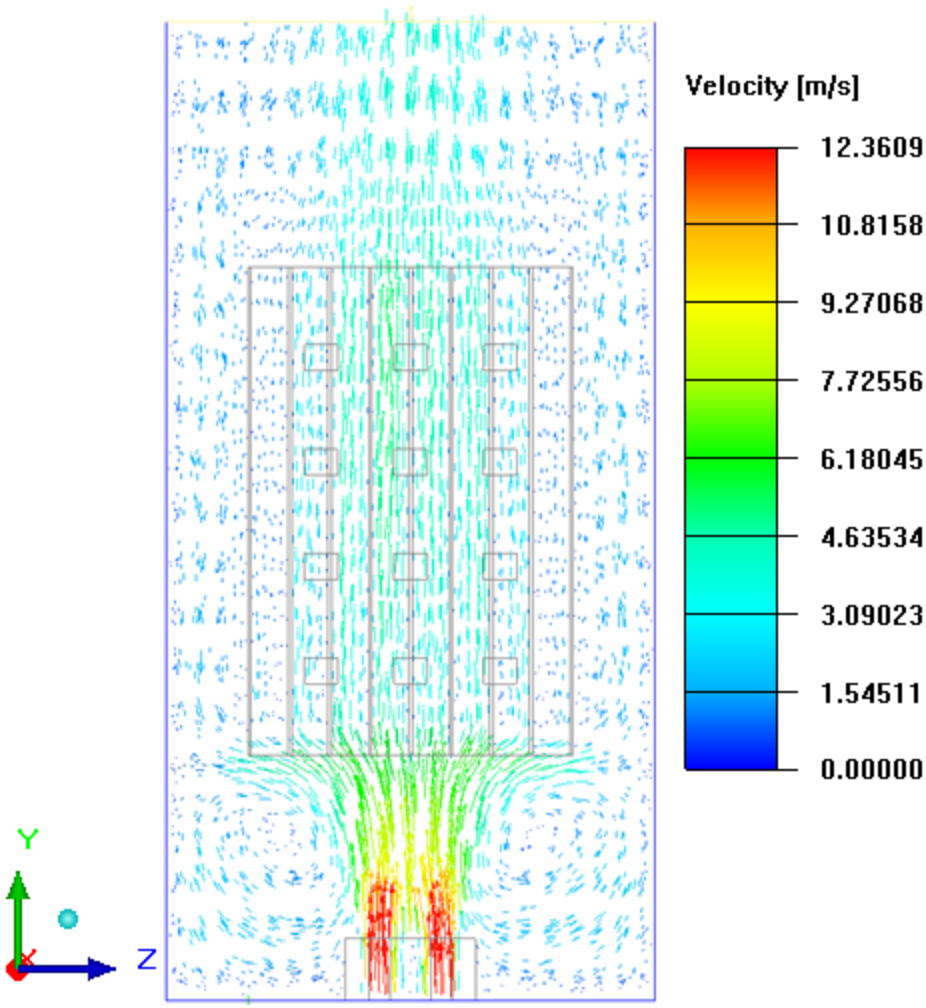
Figure 3.37: Plane Cut of Velocity Vector Field



Note:

Examining the vector plot, we can see that the flow pattern is symmetric, with two large recirculating zones adjacent to the fan. Zoom into the region directly in front of the fan and notice that two smaller recirculating zones exist in front of the hub. These local effects can be important when objects are close to the hub region.

Note:

You can move a plane cut through a model by pressing the **Shift** key, holding down the middle mouse button on an edge of the plane cut and dragging the plane cut through the model in the graphics window.

4. Create isosurfaces.

   a. Click the **Isosurface** icon ( ) in the shortcut toolbar.

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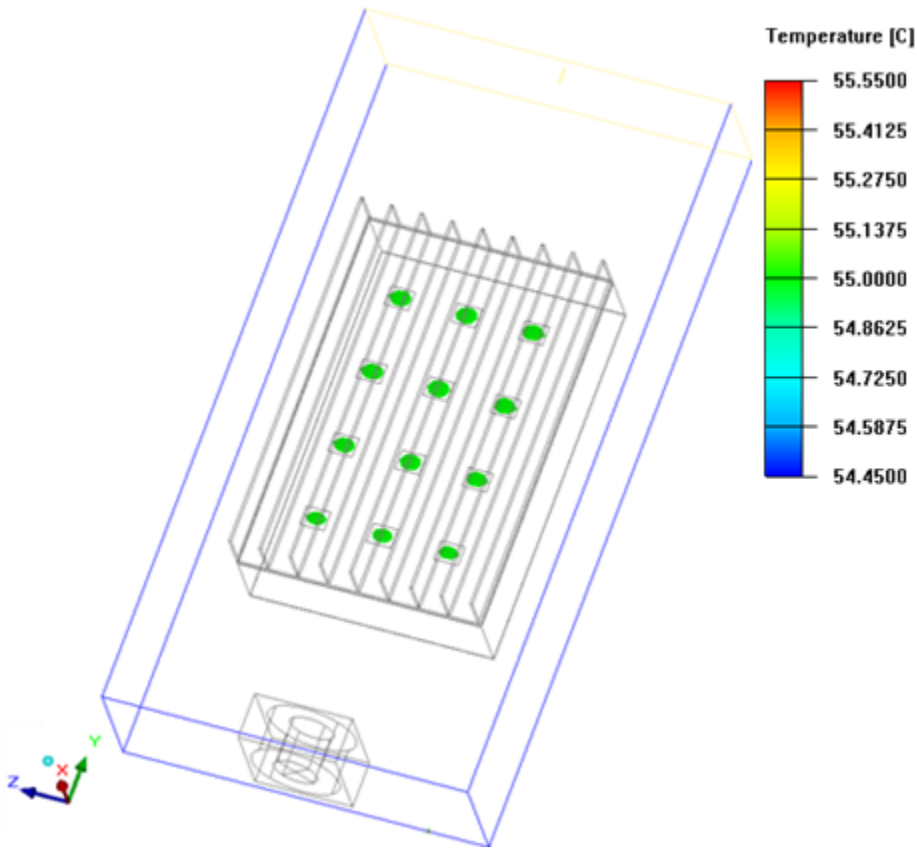
b. Specify **Temperature** as the **Variable**, input a **Value** of 55°C, and select **Show contours** and click **Parameters**. In the **Isosurface contours** panel, select **Smooth** for **Shading options** and **This object** in the drop-box adjacent to **Calculated**. Click **Done**.

c. Click **Update** in the **Isosurface** panel and notice that an isosurface has been placed around all of the sources, indicating that they have temperatures equal to 55°C (Figure 3.38: Temperature Isosurface Contour of 55°C for Sources (p. 94)).

Figure 3.38: Temperature Isosurface Contour of 55°C for Sources



d. Now, change the **Variable** to **Speed** and input a **Value** of 4. Click **Update**. Notice that the regions with velocities in excess of 4 m/s are now displayed (Figure 3.39: Temperature Contours on Isosurface of Speed 4 m/s (p. 95)).

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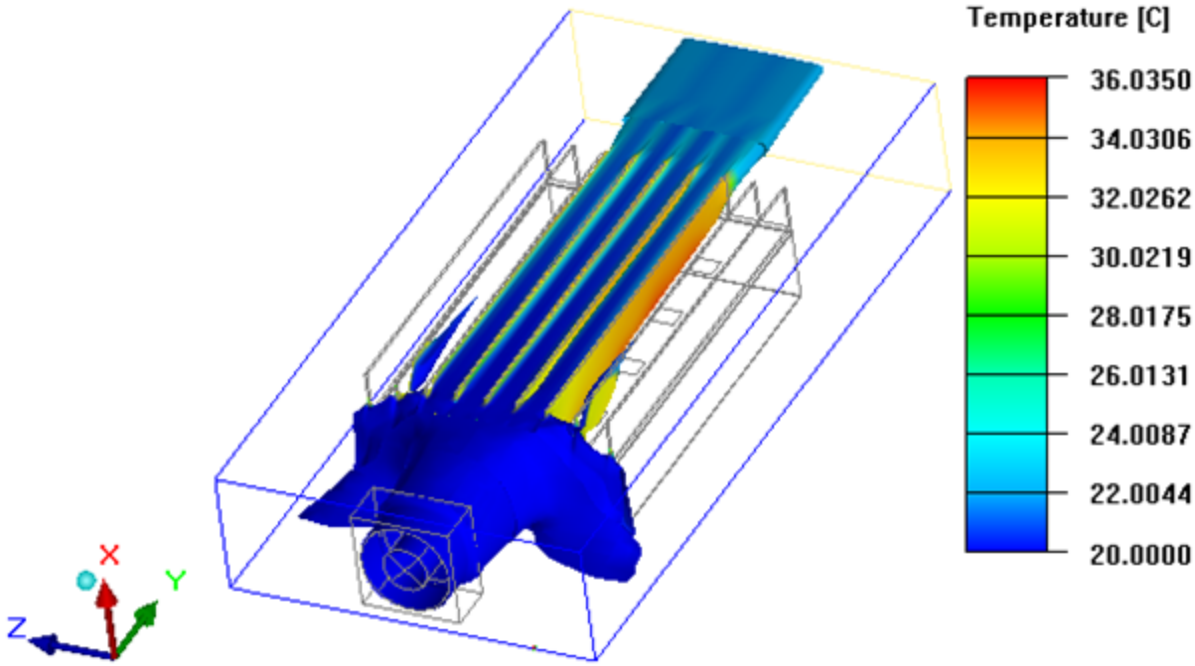
Figure 3.39: Temperature Contours on Isosurface of Speed 4 m/s

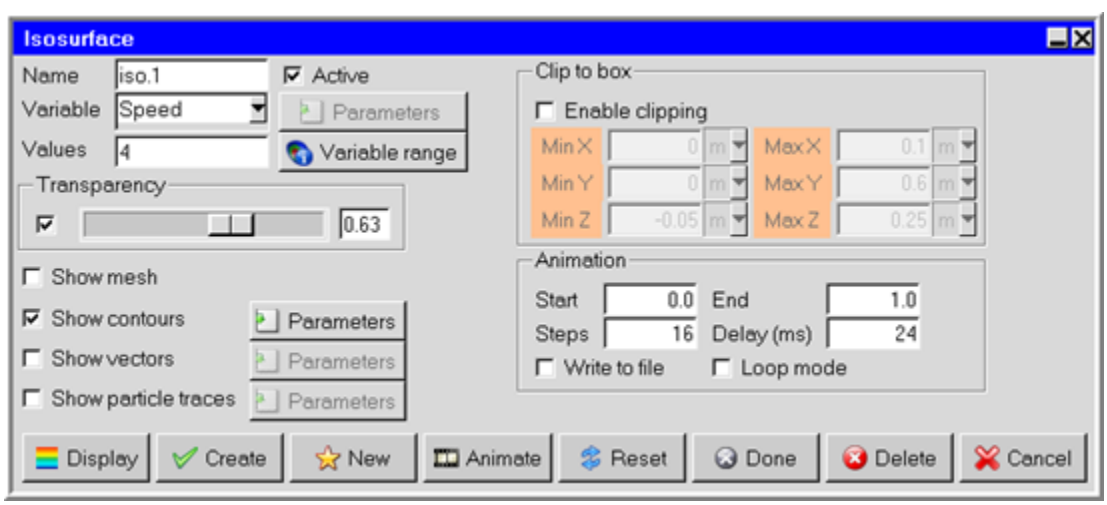


5. Utilize the **Transparency** slider to help visualize multiple post-processing objects simultaneously.

   a. With the previously created isosurface still active, select the plane cut (`cut.1`) in the **Model manager** window and select **Active** to make it visible again.

   b. Edit `iso.1`. Select the **Transparency** check box, shown in Figure 3.40: Isosurface Panel with Transparency Enabled (p. 95). Set the value to **`0.63`**.

   Figure 3.40: Isosurface Panel with Transparency Enabled



   c. Click **Done** to show the isosurface and plane cut simultaneously (Figure 3.41: Concurrent Visualization of Semi-Transparent Isosurface and Opaque Plane cut (p. 96)). Notice the semi-transparency of the isosurface allows the concurrent visualization of the opaque velocity flow field near the fan.

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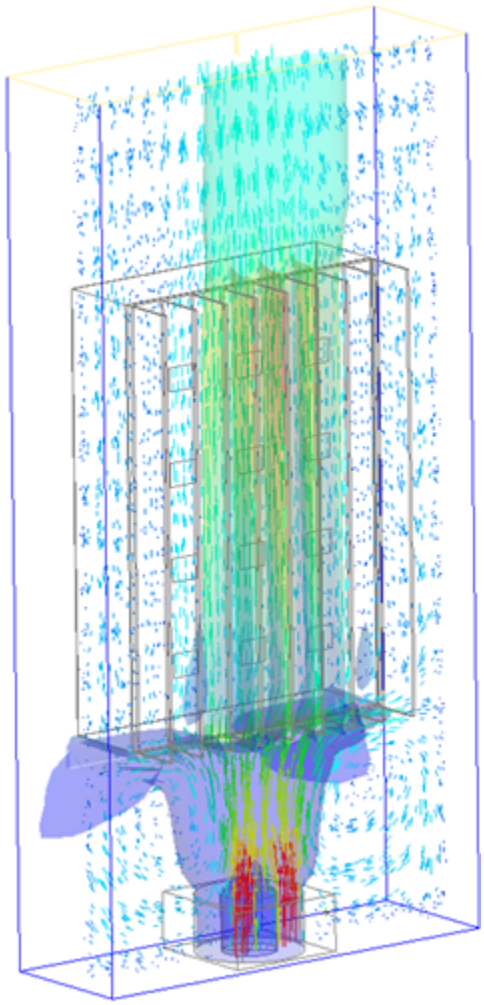
Figure 3.41: Concurrent Visualization of Semi-Transparent Isosurface and Opaque Plane cut



Note:

You can access the **Transparency** slider in the following post-processing objects: **Object face**, **Plane cut**, and **Isosurface**.

6. Create a summary report of object-specific solution data. **Summary reports** can provide physical information from the solution about specific **Model** objects, **Groups** objects, **Post-processing** objects and **Points** objects. Follow the steps below to create a summary report:

   a. First, make the post-processing object **face.1** active again by accessing the context menu under the **Inactive** node in the **Model manager** window.

   b. **Report** → **Summary report**

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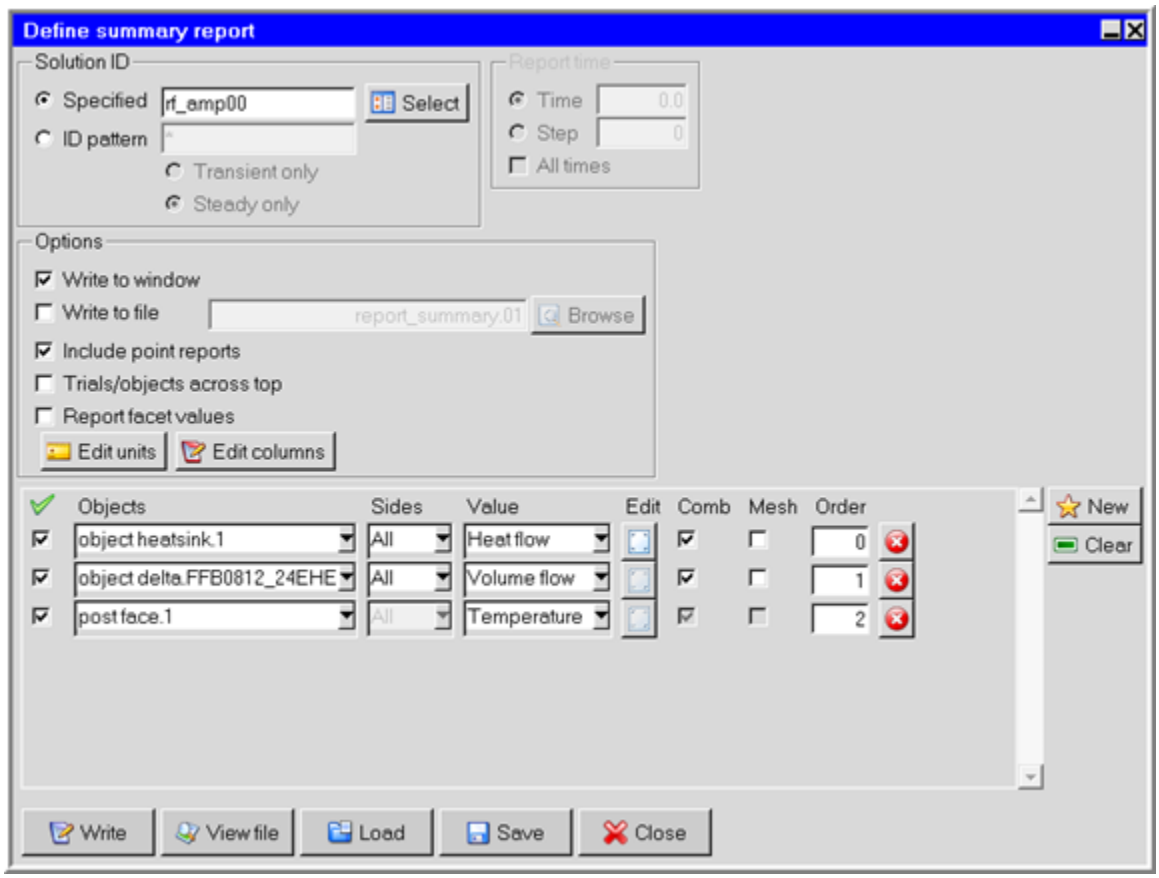

c. Click **New** in the **Define summary report** panel three times to create 3 rows of **Objects**.

d. In the first row, select **object heatsink.1**, then click **Accept**. In the **Value** drop-down menu, select **Heat flow**.

Note:

Although you did not change the default settings of the check boxes **Comb** and **Mesh**, you may encounter projects in which you need to change these. **Comb** refers to "combined": The report generates a single, combined value for all the sides selected; deselecting the option would report the side values separately. **Mesh** allows you to report on the reduced mesh of the selected object in the case that an object intersects with other objects and the mesh in the intersecting region might not necessarily belong to the object of interest.

---

e. In the second row, select the fan, **object delta.FFB0812_24EHE**, then click **Accept**. In the **Value** drop-down menu, select **Volume flow**.

f. In the third row, select **post face.1**, then click **Accept**. In the **Value** drop-down menu, select **Temperature**.

g. Click **Write** to generate a panel for the object summary report (Figure 3.42: Report summary data Panel (p. 98)). Examine the values reported and confirm they are consistent with the physics of the model. Click **Done** to exit out of this panel, then **Close** to exit the **Define summary report** panel.

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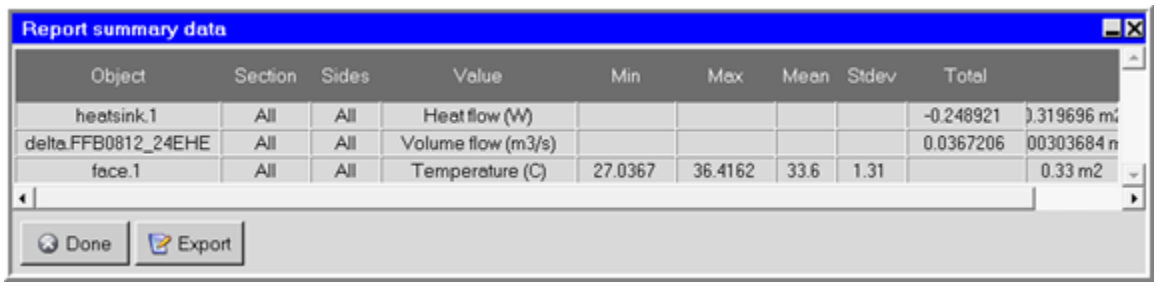
Figure 3.42: Report summary data Panel



7. Create variation plots.

   a. Click the **Variation plot** icon ( ) in the shortcut toolbar.

   > **Note:**
   >
   > Before creating the variation plots, ensure that the amplifier assembly is expanded, so that the fins are visible. Next, press **Shift**+**Z** to orient the view in the **positive Z** direction.

   b. Within the variation plot dialog box, complete the following:

      i. Specify the **Variable** as **UY**.

      ii. Click the **From screen** button.

      iii. Click the center of the heat sink fins.

      iv. Click **Create**.

   c. An X-Y plot of UY velocity versus Z coordinate is now visible (Figure 3.43: Variation Plot of UY Velocity Versus Z Coordinate (p. 99)). Toggle the **Symbols** button and notice that the velocity profile across the solution domain is represented with a solid line at the post-processing locations. Notice that Ansys Icepak creates a locally colored line according to the UY velocity magnitude, seen in the graphics window.

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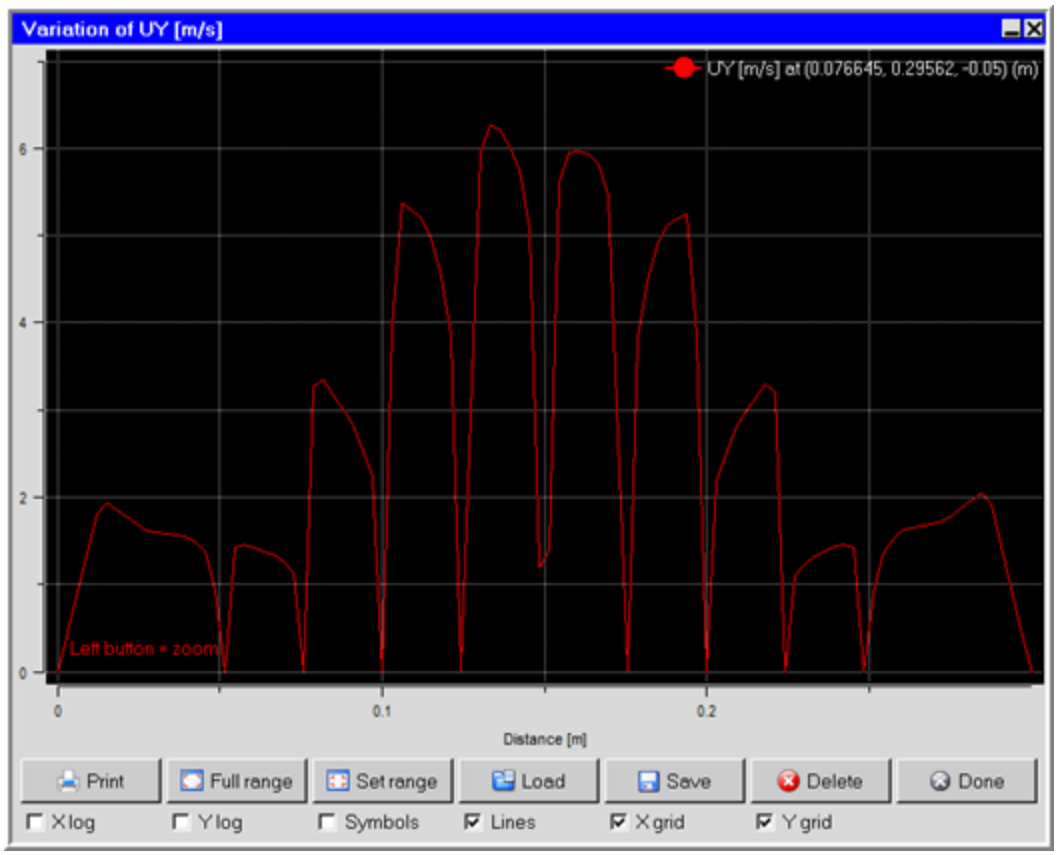
Figure 3.43: Variation Plot of UY Velocity Versus Z Coordinate



d. Save the X-Y plot.

   i. Click the **Save** button at the bottom of the **Variation of UY** plot window.

   ii. Enter a file name in the resulting **Save curve** dialog box.

   iii. Click **Save** to save the file in the model folder.

3.12. Step 9: Summary

In this tutorial, you have learned about the basic usage of enclosure, PCB, source and heat sink objects to create an RF amplifier. You have also learned how to use Ansys Icepak's fan library and search tool. Furthermore, this tutorial has introduced you to evaluating mesh quality as well as non-conformal meshing to reduce cell count and computational cost.

You have now learned the basic workflow of an Ansys Icepak project, including model building, mesh generation, problem setup, solution calculation, and post-processing as well essential features and functions that you will likely use in later tutorials or your own projects.

We repeat some of the tips and best practices found in this tutorial for your convenience:

1. Best Practices

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a. Check the default materials, properties, and operating conditions under the **Defaults** tab in the **Basic parameters** panel. These defaults often render some specifications unnecessary.

b. Use the alignment and morph tools to position objects based on other existing objects. This can be faster than specifying object dimensions and coordinates in many situations.

c. Use the **Show objects by type** feature to verify your model objects by type.

d. View the HTML summary report (**View** → **Summary (HTML)**) to ensure proper specification of geometries, properties, and materials for each object.

e. Reduce mesh counts and consequently decrease run times in regions requiring less resolution by creating separately meshed assemblies when appropriate. Also select suitable slack values that improve the convergence rate while avoiding mesh bleeding.

f. Select the **Allow minimum gap changes** option in the **Misc** tab of the **Mesh control** panel to allow Ansys Icepak to avoid unnecessary meshing due to inadvertent misalignments in the model. This is suitable for this tutorial but may not be in other projects.

g. Ensure that you have at least three to four cells between fins to resolve adequately the boundary layers between them by visually inspecting the mesh. Refine the mesh as necessary by using **Normal** meshing.

h. Evaluate your mesh quality under the **Quality** tab in the **Mesh control** panel.

i. Use the **Problem setup wizard** for guided problem setup. Edit the problem setup if needed using the **Basic parameters** panel.

j. Judge convergence by monitoring residual levels as well as relevant integrated quantities for certain objects with point monitors, which should eventually stop changing significantly with more iterations.

2. Tips and Tricks

a. Explore the several methods to complete a task as you work through a problem in Ansys Icepak. For example, you can edit model object geometries by using their object edit panel, or you can use the **Edit** window in the bottom right-hand corner while having the object node selected in the **Model manager** window. Use the options best suited for you to help streamline your workflow.

b. Cycle through all possible edges in the **Morph edges** mode by repeatedly clicking the left mouse button. This is useful in case you have difficulty selecting an edge.

c. Remove object names by clicking the **Display object names** button ( ) to reduce visual clutter in the graphics window.

d. Highlight critical regions using the **Power and temperature limits** feature (**Model** → **Power and temperature limits**) to designate maximum power and temperature, which can then be highlighted in the graphics window after the solution calculation (**Post** → **Power and temperature values**). Note that setting these limits does not affect how Ansys Icepak calculates the solution.

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e. Use the **Reset** button in the **Basic settings** panel to have Ansys Icepak estimate dimensionless numbers (for example, the Reynolds and Peclet numbers for forced convection) and determine the appropriate flow regime.

f. Hold your mouse pointer over any selection in the **Problem setup wizard** to have a text bubble appear for additional information on the selection.

g. Use the **Transparency** slider in the post-processing object edit panels to improve visualization of post-processing objects.

h. Move a plane cut through a model by pressing the **Shift** key, holding down the middle mouse button on an edge of the plane cut and dragging the plane cut through the model in the graphics window.

i. Expand all of your assemblies quickly by right-clicking **Model** in the **Model manager** window then selecting **Expand all**.

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